

SERVICE APPLICATIONS

Introduction

The Service Application gives Service the ability to easily set up the AcQSim CT system via a Graphical User Interface (GUI). The different setup menus include General Utilities, DITA, Calibrate/Perf. and Configure.

Service Applications Procedure

1. Boot the system. Select the SERVICE button from the opening screen. Refer to Figure 352. SEE NOTE below:

NOTE

The [Diagnostic GUI](#), which initiates AcQImage Diagnostics, Gantry and other utilities, may be started at this time (before starting the Service App) by clicking in the bottom left corner (slightly smaller than 1" square) and typing **PICKER** <CR>. After exiting the Diagnostic GUI, the Service Application Selection screen will return (Figure 352).

Click once in the bottom left corner. Type PICKER (caps) and press <CR> to activate the Diagnostic GUI.

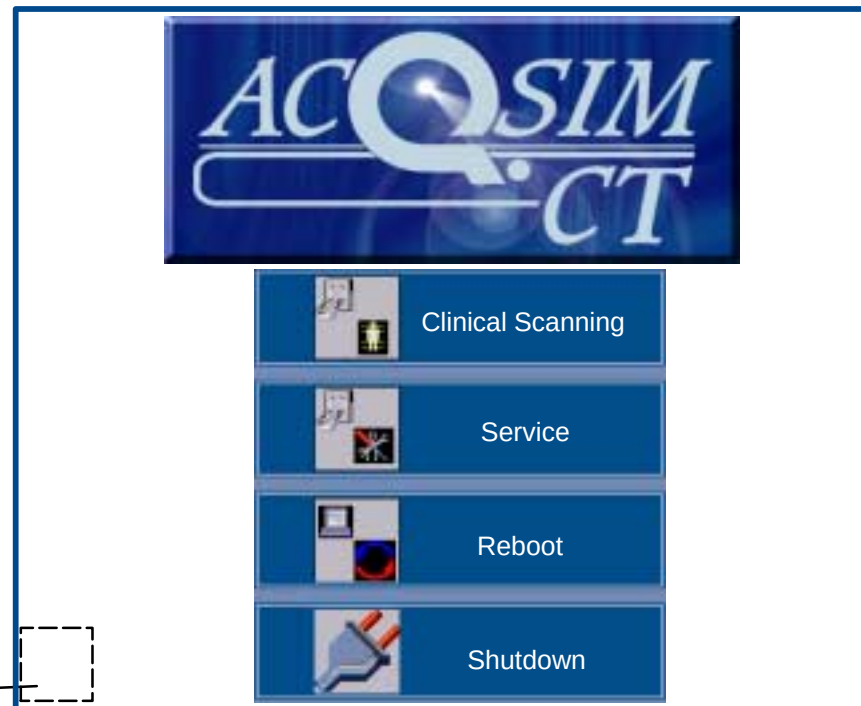


FIGURE 352 SERVICE APPLICATIONS SELECTION SCREEN

2. The opening screen defaults to “General Utilities”: At the bottom of the screen are the four major categories of Service Applications:

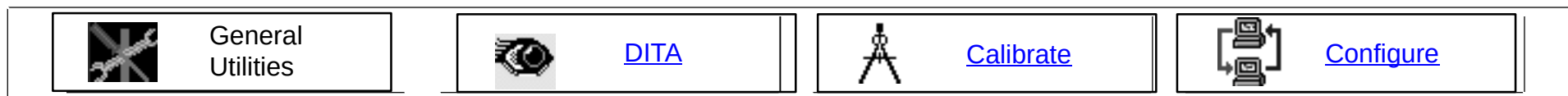


FIGURE 353 SERVICE APPLICATIONS – MAIN CATEGORIES

The following procedures detail each individual Service Application, based on Figure 353, going from left to right (General Utilities | DITA | Calibrate | Configure). Click on the links above for more information.

GENERAL UTILITIES

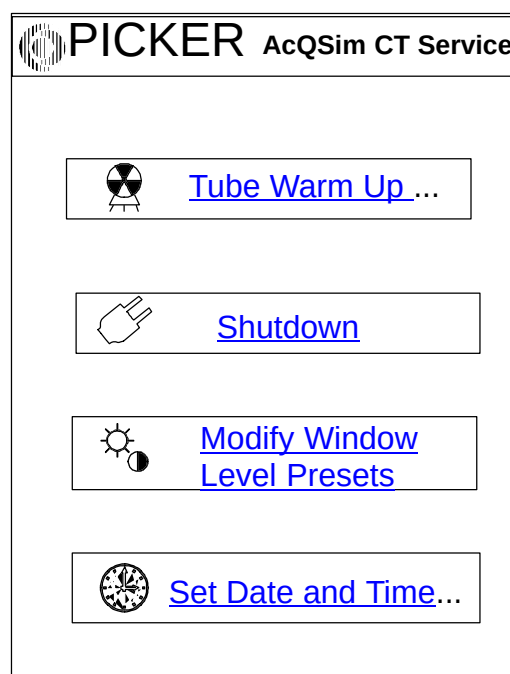


FIGURE 354 GENERAL UTILITIES WINDOW

The **General Utilities** menu includes the following options. Click on the respective buttons in Figure 354 for more information.

Tube Warm Up: Ensures proper tube heat necessary before taking a scan. Press the Start Scan button to begin warmup. Status is seen in the Progress and Scan Status windows.

Shutdown: Used to start/shutdown Gantry, Console –(Display Tower and AcQImage Cabinet) or shut down the entire system. Switch to Console launches the Console App.

NOTE: If the Gantry power is on, the third button will read “Shutdown Gantry.” If the Gantry power is off, the button will read “Start Gantry.”

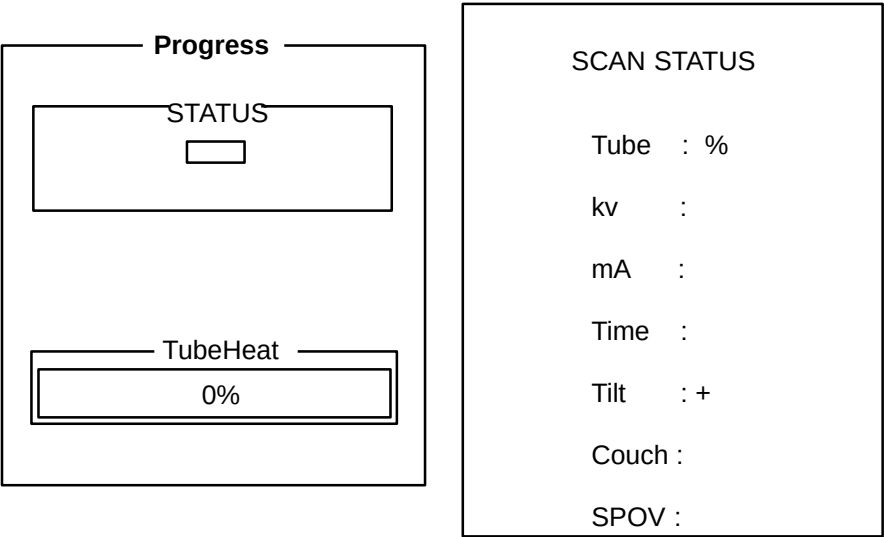


FIGURE 355 TUBE WARM UP

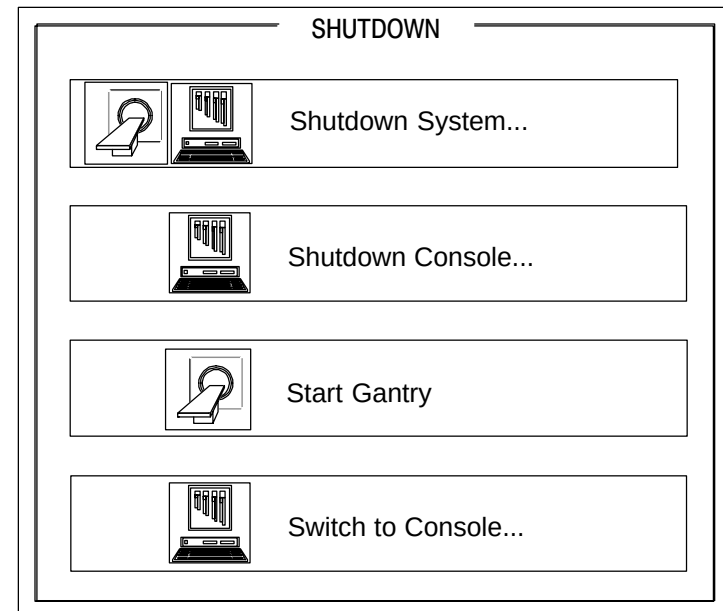


FIGURE 356 SHUTDOWN SCREEN

Modify Presets:

Opens the Window/Level Presets Editor. You may add customized presets or delete existing ones. To add, click on the ADD button to highlight the next preset row. Click in the cell under the Preset column and type in a preset name. Repeat this in the window and level cells and click on SAVE. To delete, click on the preset value, then click on the DELETE button. If you make a mistake, click on the CANCEL button.

WINDOW LEVEL PRESETS EDITOR		
Preset	Window Width	Window Level
Brain	180	45
Mediast.	400	20
Spine	350	70
Abdomen	350	50
Lung	1600	-300
Bone	2000	350
Pilot	800	500

FIGURE 357 MODIFY PRESETS

Set Time and Date: Allows you to set the time and date of the system, time zone, and 12/24 hours clock. Use either up/down arrows, toggle buttons or pulldown menus. When the time settings are correct, click on the SET TIME button. When the date settings are correct, click on the SET DATE button.

TIME

04:45 PM

Hour

MIN

AM

PM

TIMEZONE: US/Eastern

12 HOUR

24 HOUR

Set Time

Reset

DATE

APRIL 1998

S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

Set Date

Reset

DONE

FIGURE 358 SET TIME and DATE

Philips Medical Systems. Confidential and Proprietary information. Refer to Title Page.

DITA

The AcQSim CT system supports the storage and display of data at many steps of the reconstruction process. A utility called DITA is used to display this information and can be used to isolate faults ranging from individual detector channels to failure of entire processing section such as the AP board. DITA opens with the study selection screen:

Select Raw Data							
Study #	Series #	Patient	Name	Date	Time	Scan Time	Data Type

SEARCH

Entry:

Next

Previous

SERIES INFORMATION

First Name:
Last Name:
Reference:
Protocol:

Study No.:
Scan Type:
Data Type:
No. of Revs:

Acq.Data/Time:
Recon. Date/Time

Start Couch Pos.:
End Couch Pos.
Data Location

Filter Raw Data by Type

☒ Detector Fan

Select All

Clear All

FIGURE 359 OPEN STUDY LISTING

Corresponding data will appear in the menu selection list. Highlight the study of choice, then click on OK.

Measure Selection Window

After the data loads, it will be displayed in the right window. The upper left window has three options and defaults to FILM, which would be used if filming the image was desired.

Click on the MEASURE tab. The Measure Selection box will appear:

Display Mode Selector: Choose the way you want the data to be displayed: Integer, Hex or Volts.

FILMMEASUREBAD DETS

Hints

Press Stop Measurement to cancel

Measurement

[Report Cursor](#)

[Data Profile](#)

Display Mode

INT☐

Cancel

Stop Measurement

Clear Last

Clear All

Pre Data Words

Updated By 'Report Cursor' Measurement

View Number

View Status

Revolution No.

Couch Position (mm):

*

*

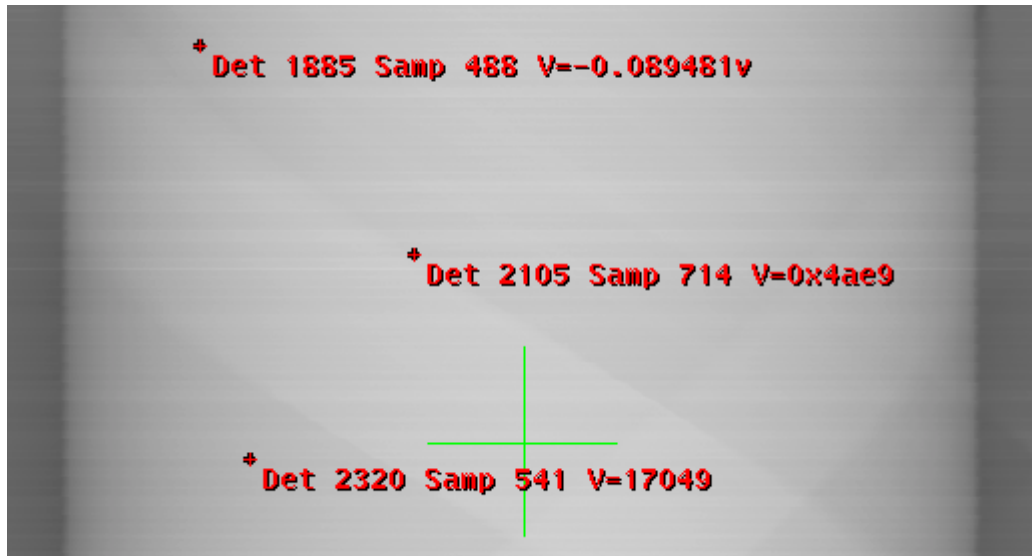
*

*

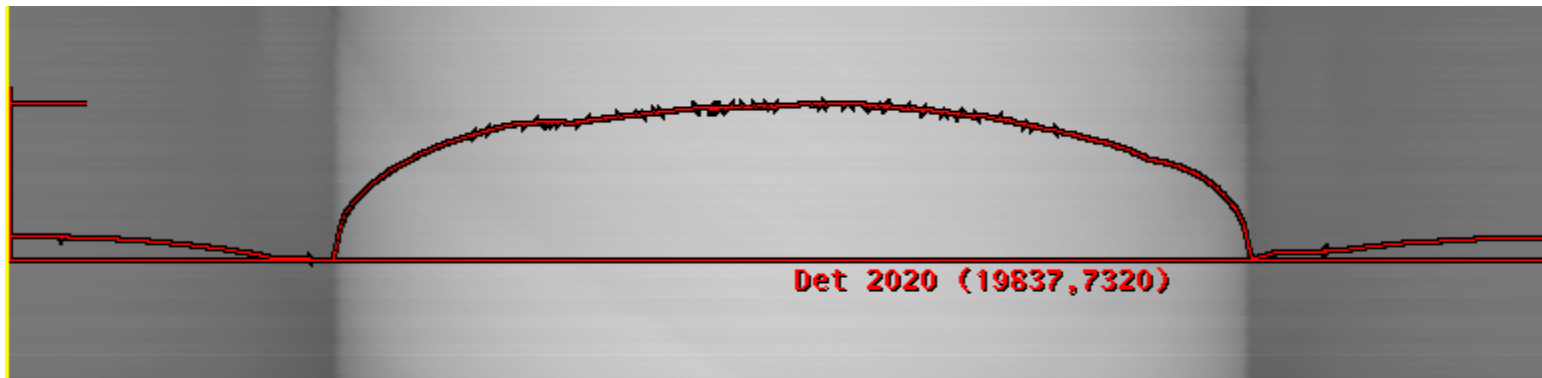
FIGURE 360 MEASURE WINDOW

The MEASURE SELECTION BOX contains the following measurement selections:

REPORT CURSOR: When the Report Cursor from Figure 360 is selected, you can move the cursor to the right window where the data is displayed. The cursor will change to look like a pencil point. This cursor is used to select an individual detector or area of the data. When you select a point on the data, a short report will appear. In this example, Detectors 1885, 2105 and 2320 were selected:

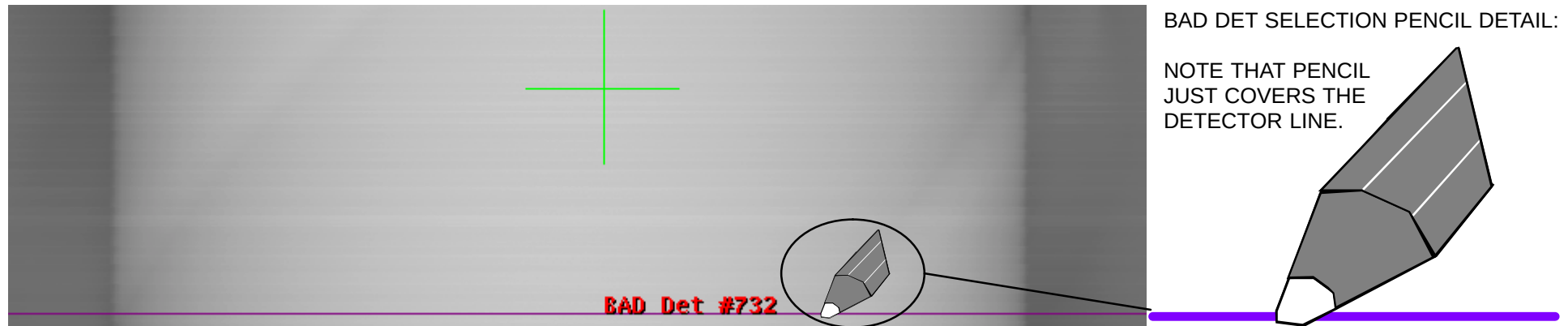


DATA PROFILE: When the Data Profile button in Figure 360 is selected, you can select an area of data that you want a profile on using the pencil point cursor. In this example, detector 2020 was selected:



Bad Dets Selection Window

To review the bad detector list, or add/remove a bad detector from the list, click on the BAD DETS tab. The Bad Dets selection window will be displayed. If you select the “Highlight Bad Dets” button, any listed bad detector will appear on the image data:



The Bad Detector List will update immediately after a detector is added or removed from the list. If changes are made to the list and you do not save it, you will be prompted when exiting DITA (see Figure 363).

FILM	MEASURE	BAD DETS
------	---------	-----------------

☐ Highlight Bad Dets

Add or Remove a Bad Detector

Add: Remove:

Restore Original Bad Dets

Bad Detector List

732

Save
Bad Detector List

FIGURE 361 BAD DETS WINDOW

The bottom left window contains AcQquisition Information as seen in Figure 362.

Enter the Detector Number to move the displayed data within the window.

Pan and Zoom: Grab the purple box within the display window. Move the box around to pan, increase the size of the box by pulling one of the four corner “handles” outward to zoom.

RAW

Acquisition Selection

Total Acqs: 12

Acq: 1

Total Revs: 1

Revs Loaded: 1 -1

Rev: 1

Detector: 392

Sample: 679

Zoom: 10.0

Window/Level

W: 1000

L: 0

FIGURE 362 RAW DATA INFORMATION WINDOW

If any changes to the bad detector list were made and not saved, the following message will appear before the program terminates:

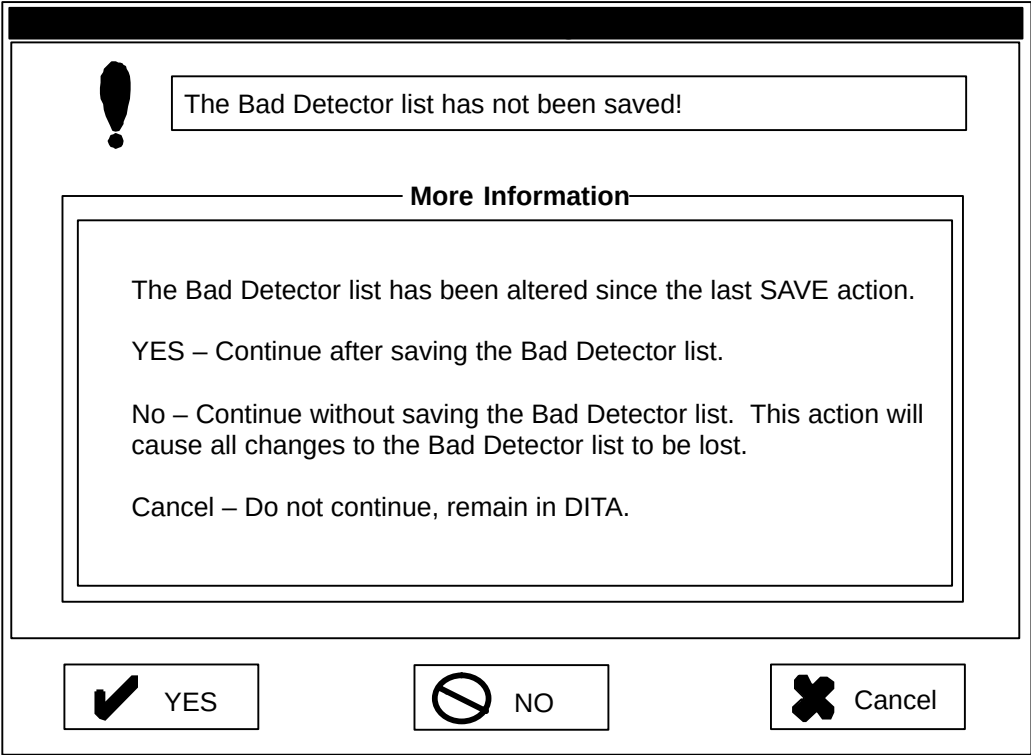


FIGURE 363 EXIT WINDOW

AUTO-CALIBRATION

Introduction

The **calibration** worksteps provide a way to calibrate the AcQSim CT scanner. An automatic sequence will control couch movement, scanning with desired parameters, and running a desired analysis program. It also implements a **performance** assessment sub-system for determining the relative performance of a CT system.

The **User Interface** is no longer via the ELTP like the Q series scanners. All calibration workstep interfaces are GUI based, (Graphical User Interface) with selections made via a mouse “click.” Each GUI has a cohesive purpose and is described below.

Calibration makes **Automatic Bad Detector Entries** in the vector database. Calibration will write a list that fails to meet spec (see below). Calibration will not remove any detectors from the bad list. Manual entry and removal of bad detectors is still available via DITA. Detector status is also available via DITA.

FIELD DETECTOR MODULE REJECTION CRITERIA

- a. There can be no more than 16 bad detectors per system / else replace the module with the most bad detectors.
- b. There can be no more than 3 bad channels per detector module / else replace the module.
- c. There can be no more than 2 adjacent bad detectors, either in a module or in adjacent modules / else replace the module.
- d. No defective channel can adversely effect other channels in a quadrature manner / else replace the defective component causing the adverse effect.
- e. Any module with a defective ARDAS channel must be replaced.

Please promptly return the modules for repair with a description of the defects, date and scanner I.D. Knowledge gained from any reported defects will be applied to reduce the number of defects in future production.

The next several pages describe the GUI workstep interfaces. The first graphic shows the entire Calibration GUI to visualize where there individual workstep GUI areas are located. The are detailed in subsequent graphics.

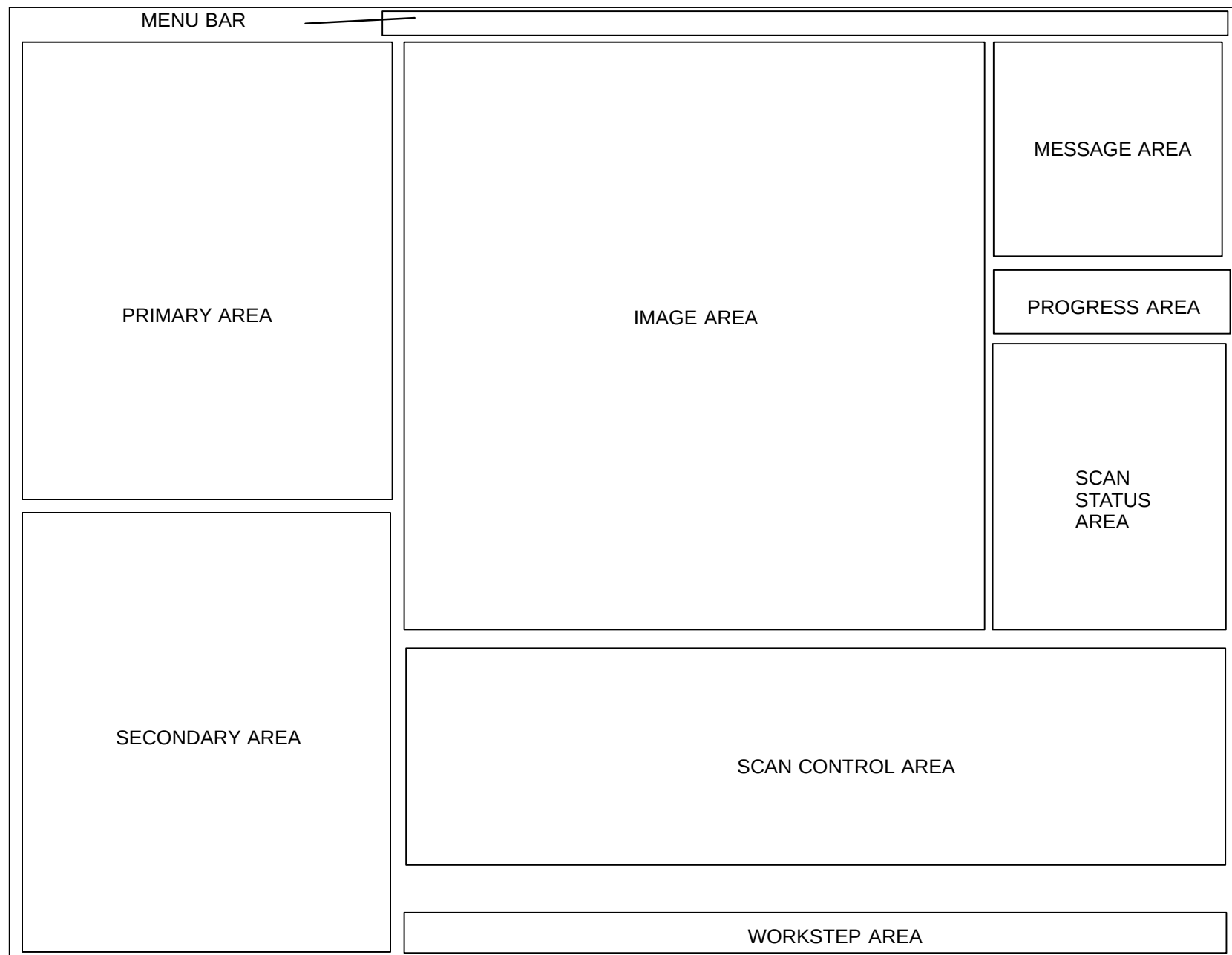
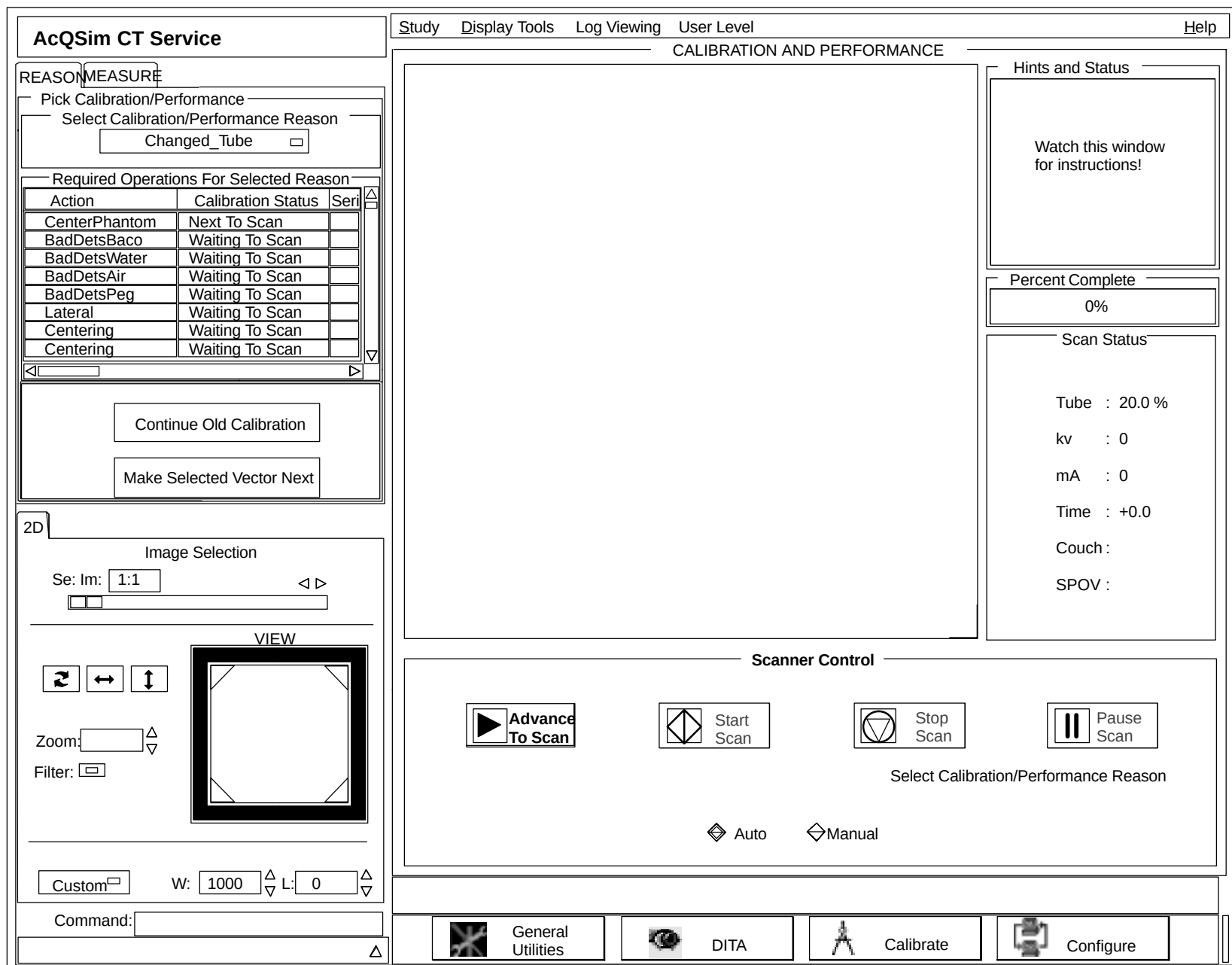


FIGURE 364 CALIBRATION WORKSTEP GUI LAYOUT – FULL SCREEN (NO DETAIL)



PL0078

FIGURE 365 CALIBRATION WORKSTEP – FULL SCREEN (WITH DETAIL)

PRIMARY GUI AREA

The **PRIMARY AREA** contains all "primary" functions. The calibration workstep divides this area into three major functions: REASON, MEASURE and FILM. Refer to Figure 366.

Reason GUI (Tab 1):The reason gui allows the user to select the reason for calibrating or doing performance of the system. The gui has two main parts: the Reason Pull-Down Menu and the Vector Grid:

Reason Pull-Down Menu: This menu lists all possible reasons for calibrating. When starting a calibration or performance the user simply selects one of these reasons. The calibration workstep will determine all calibration or performance functions required by this reason. It will fill in the Vector Grid accordingly and then enable the advance to scan button of the X-ray Control GUI.

REASONMEASURE

Select Calibration / Performance

Select Calibration / Performance Reason

See Reason Chart

Required Operations For Selected Reason

Action	Calibration Status	Seri
BadDetsBaco	Scanning	
BadDetsWater	Next to Scan	
BadDetsAir	Waiting To Scan	
BadDetsAir	Waiting To Scan	
BadDetsPeg	Waiting To Scan	
Lateral	Waiting To Scan	
Centering	Waiting To Scan	
Lump	Waiting To Scan	

Continue Old Calibration

Make Selected Vector Next

FIGURE 366 PRIMARY GUI AREA

Vector Grid: The vector grid shows one row for each vector that will be created or SPEC (Scanner Performance, Evaluation, and Correction) action to be taken. Each row lists the action, an event number, status and the Vector Grid parameters for creating that vector: KV, MA, etc. The user may (not recommended) start the calibration at any of these rows (vectors or actions) by simply double-clicking anywhere in that row. The status of each vector, as shown in column three of the vector grid, is very important when problems arise. You can watch the normal state of progression: Waiting to Scan->Next to Scan->Scanning->Calculating->Complete

There are other states that may appear FOR A SHORT TIME. Any other vector state which remains for longer than a minute indicates a problem. In manual mode the states are different.

Measure GUI (Tab 2): The measure gui operates the same as in the scanning workstep and thus will only work when an image is on the screen and preview is off.

Film GUI (Tab 3): Works the same as the scanning workstep.

Reason Pull Down Menu

REASON	DESCRIPTION OF ACTION PERFORMED
Changed_Tube	
Changed_Detector_Module	
Changed_VFSCCard	
Changed_HighVltgeModule	
Changed_Collimator_Module	
Changed_Patient_Aperature	
Changed/Moved_Motherboard	
Changed_XSC	
Verify_Gantry	
Check_Performance	
Center_Phantom	
Custom_MTF	
Custom_Scan	
Clear_BadDetsList	
Find_BadDets	
Perform_Lateral_Centering	
Find_Centering	
Find_DAC	
Find_Air	
Find_MA	

Find_PilotAir	
Find_Flatts	
Find_Level	
Find_Level2	
TouchUp_Air	
Perform_MTF	
Perform_KVHVL	
Perform_LowCon	
Perform_CNA	
Perform_Uniformity	
Perform_SliceThickness	
Evaluate_ARDas	
ManBadDets_Baco	
ManBadDets_Water	
ManBadDets_Air	
ManBadDets_Peg	
ManPerform_Lateral_Centering	
ManFind_Centering	
ManFind_DAC	
ManFind_Air	
ManFind_MA	
ManFind_PilotAir	
ManFind_FlattsFF	
ManFind_LevelFF	
ManFind_Level2FF	
ManPerform_MTF	
ManPerform_KVHVL1	
ManPerform_KVHVL2	
ManPerform_LowCon	
ManPerform_CNABaco	
ManPerform_CNAWater	
ManPerform_CNAResolver	
ManPerform_UniformityFF	
ManPerform_SliceThickness	

SECONDARY GUI AREA

The **Secondary Area** contains functions considered secondary:

2D View GUI: Works the same as scanning and will only be visible if there is an image on the screen. Some functions only work when preview is off such as positioning to a new series.

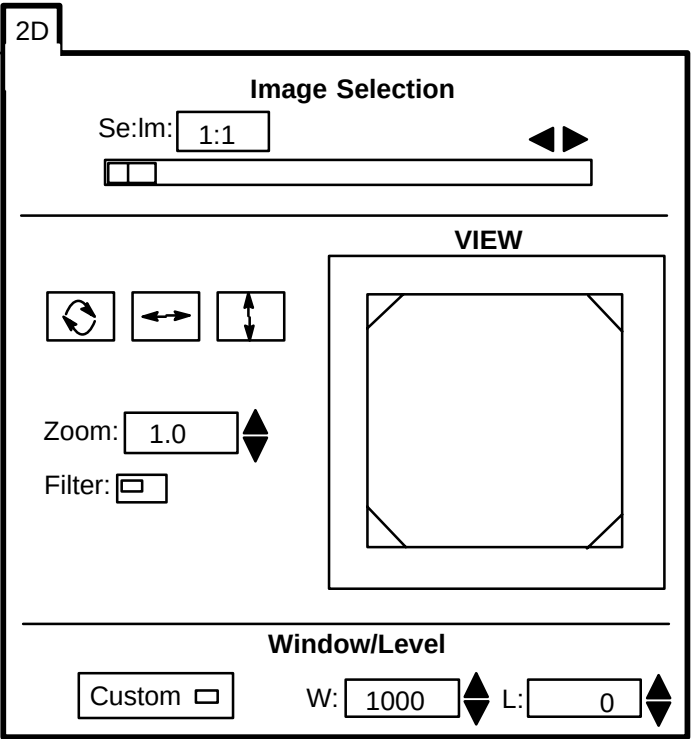


FIGURE 367 SECONDARY GUI

WORKSTEP FLOW AREA

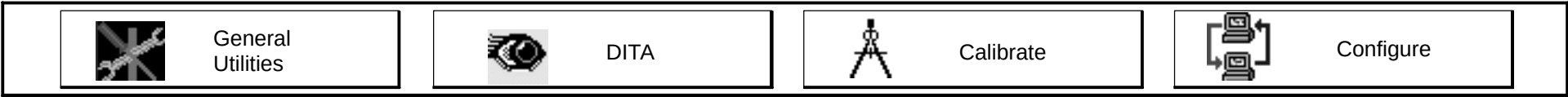


FIGURE 368 WORKSTEP FLOW AREA GUI

The **Workstep Flow area** contains the workstep flow buttons. While in the calibration workstep, if calibration is active in any way you **will not** be allowed to leave the calibration workstep until all activity has ended.

The **PREVIEW GUI** (Not shown yet) turns previewing of images ON or OFF.

SCANNER CONTROL AREA

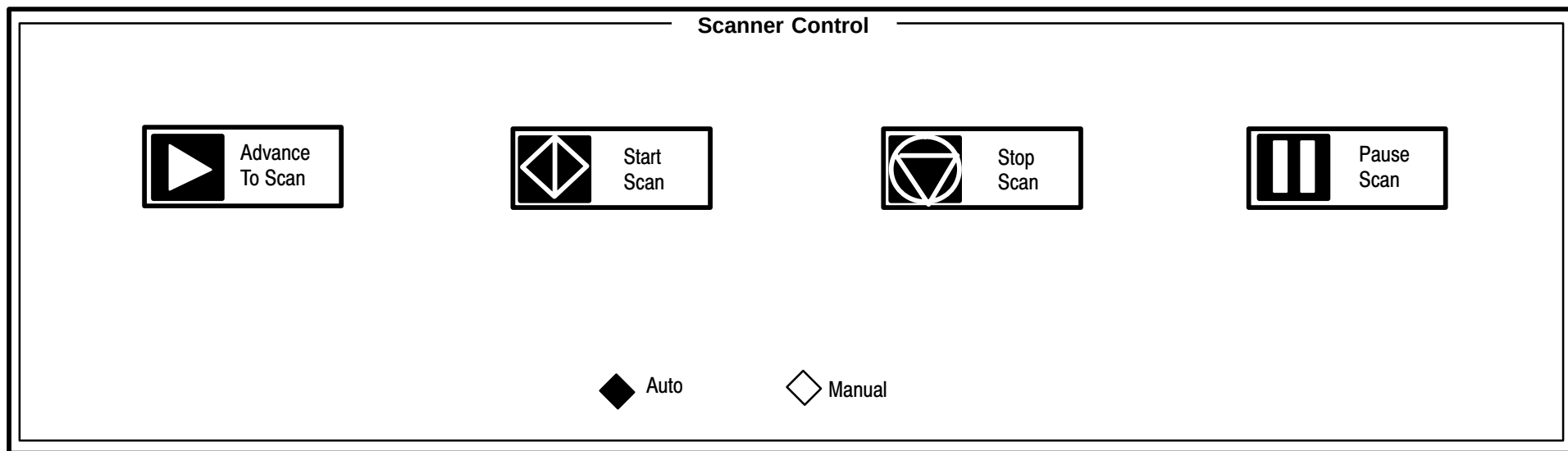


FIGURE 369 SCANNER CONTROL AREA GUI

The **Scanner Control area** has all scanner control buttons which are described relative to calibration in the next five sections. They include Advance to Scan, Start Scan, Stop Scan and Pause Scan:

NOTE

The buttons gray out when their meaning is no longer valid for the calibration workstep's state. Thus, the pause button is grayed out (inactive) if no reason has been selected.

Advance To Scan: Is enabled once a reason has been asserted and before calibration scanner activity starts. When pressed will cause the scanner to spin up the rotor and some other preliminary activities which are to be determined.

Start Scan: Actually starts the calibration or performance activity by scanning to collect data. This button is enabled only when we are ready to scan.

Stop Scanning: Will cause the calibration workstep to stop scanning and stop all math processes, but continues until all queued data has been consumed. When the math server finishes its calculations, all scanning stops and all plans are stopped the advance to button is enabled and so is the selection of a new calibration reason.

Pause Scanning: Will cause calibration to stop scanning after the current scan has finished.

Auto/Manual Radio Buttons: If auto is selected calibration proceeds in it's most automatic mode. That is, minimal operator intervention is needed. If in the manual mode then the start button must be pressed after each calibration vector action is taken (when a row is finished in the grid widget).

SCAN STATUS AREA

The **Scan Status** area shows the status of the Tube heat, tube kV and mA, time elapsed, degree of gantry tilt and the focal spot.

Scan Status

Tube : 10.0%

kV : 0

mA : 0

Time : 0

Tilt : 0

SFOV : Large

FIGURE 370 SCAN STATUS AREA GUI

CALIBRATION / PERFORMANCE PROGRESS

The **Calibration / Performance Progress Area** indicates the time calibration has been active, an estimate of how much longer it will take, and the percent complete.

NOTE: Errors and pauses will affect the accuracy of this widget.

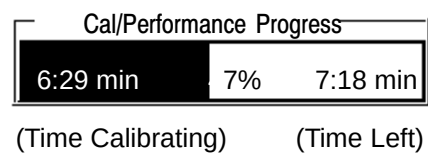


FIGURE 371 PROGRESS GUI

CALIBRATION MESSAGES

The **Calibration Messages** area displays operational messages to the user. These messages may help you determine what to do next or may simply give status.

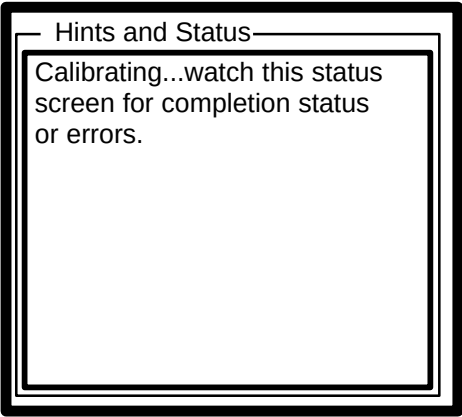


FIGURE 372 CALIBRATION MESSAGES GUI

DB GUI

Open Incomplete Calibration/Performance Study						
STUDY/SELECT				STUDY INFORMATION		
Study#	Scanner	Patient Name	Date	A	TY	Study Description
SEARCH						
Entry:		NEXT	PREVIOUS			
				First Name: Last Name: Sex: Patient Name: Comments: History: Diagnosis: Radiologist: Ref. Phys.: Time:		

FIGURE 373 DB GUI

The **DB GUI** Appears when you select the “Continue old cal or perf” button in the primary area.

You are responsible for removing the studies created by calibration. Every study created by calibration will have the last name: CALIBRATION.

CALIBRATION PROCEDURE

Calibrating the AcQSim CT system has been greatly simplified. The Workstep GUIs show most of the information needed to perform a system calibration. A basic procedure is below:

1. Select a reason from the Primary Area pull-down menu, or select “Continue old Cal or Perf.”
 - If an old study DB gui appears on screen or if an old calibration is in the DB and you select “continue”, from the DB area, then the reason selected will to be the old one.
 - If you want manual mode, select it in the Scanner Control area.

2. Click on the “Advance to Scan” button in the Scanner Control area.
3. Click on the “Start Scan” button in the Scanner Control area.
 - You may have to hit start each time if in manual mode or in lateral centering.
 - You may click on “Preview Off” to use the full features of measure gui.
4. Wait for completion of the calibration sequence, while noting the changing states in the Primary Area vector grid.

AUTOPERFORMANCE

AutoPerfromance is performed in the same manner as Autocalibration, selecting the Performance “Reason” from the REASON tab in the Primary GUI window. See the [Reason Pull Down Menu](#) Display for selections.

CONFIGURE

The different CONFIGURE menus for the SCANNER include Camera, Dicom, Network, Options, AcQImage, Boards, and Site configurations. Refer to Figure 374. UltraZ Diagnostic Workstation Configure follows.



FIGURE 374 CONFIGURATION SELECTIONS

The following procedures detail each individual configuration, based on Figure 374, going from left to right.

You may click on the buttons in Figure 374 to jump to their respective screens.

Additional information on networking terms can be found in: [Understanding IP Addresses, Subnetmasks, and Routing](#)

Site: **Allows complete SITE configuration.**

Site Configuration

Institution Name

Institution Address

Department Name

Station Name

Equipment Manufacturer

Model Name

Equipment Serial Number

Software Version

Spatial Resolution

Image Compression

Last Calibration Date

Last Calibration Time

FIGURE 375 SITE CONFIGURATION

Boards: Allows complete system BOARDS configuration.

RECONSTRUCTION BOARD CONFIGURATION

Select Board

Board Type	Serial Number	Ethernet Address	

Delete Board

Add Board

Edit Board

◆ CP

◇ GAM

◇ DSPAP

◇ PROJECTOR

Serial Number

Ethernet Address

Apply

FIGURE 376 BOARDS CONFIGURATION

BOARDS – ETHERNET AND IP ADDRESS CONFIGURATION

There are 3 ethernet networks employed within the product. Refer to Figure 377.

- Private internal ethernet network: Used for communication between boards in the ACQIMAGE CABINET. Configurations may be necessary.
- Private internal 100 Mbit ethernet network: Used for communication between the ACQIMAGE CABINET and the DISPLAY CONSOLE. No configuration required.
- Standard ethernet NETWORK PORT on the DISPLAY CONSOLE: Used for network connections to options and external hospital networks. Configuration is necessary.

NOTE

Refer to the [Networking Glossary](#) for assistance with definition of terms.

Systems from the factory will have ethernet addresses already assigned to the AcQImage Cabinet boards. If a board is added or changed in the field the new ethernet address assigned to the board(s) will need to be entered. Each board is physically labeled with a serial number and ethernet address. Values are entered in the Configure | Boards option of the Service Application. Refer to Figure 376.

Once the ethernet address(es) has(have) been updated, IP addressing for the board will automatically be generated and loaded into the configuration file as well. There is no need to establish IP addresses for the boards as they will be assigned according to a predetermined scheme. Default IP addresses will be assigned to the boards according to the order they are entered in the configuration:

192.168.2.10 – First board

192.168.2.20 – Second board

192.168.2.30 – Third board

192.168.2.40 – Fourth board, etc.

IP ADDRESSING – 100 MB ACQIMAGE / DISPLAYCONSOLE LINK

IP addresses for the 100 MB link are preset to default values by the applications software. The following addresses are assigned and need not to be changed:

192.168.1.1 DISPLAY CONSOLE 100 MB port,

192.168.1.2 ACQIMAGE CABINET 100MB port

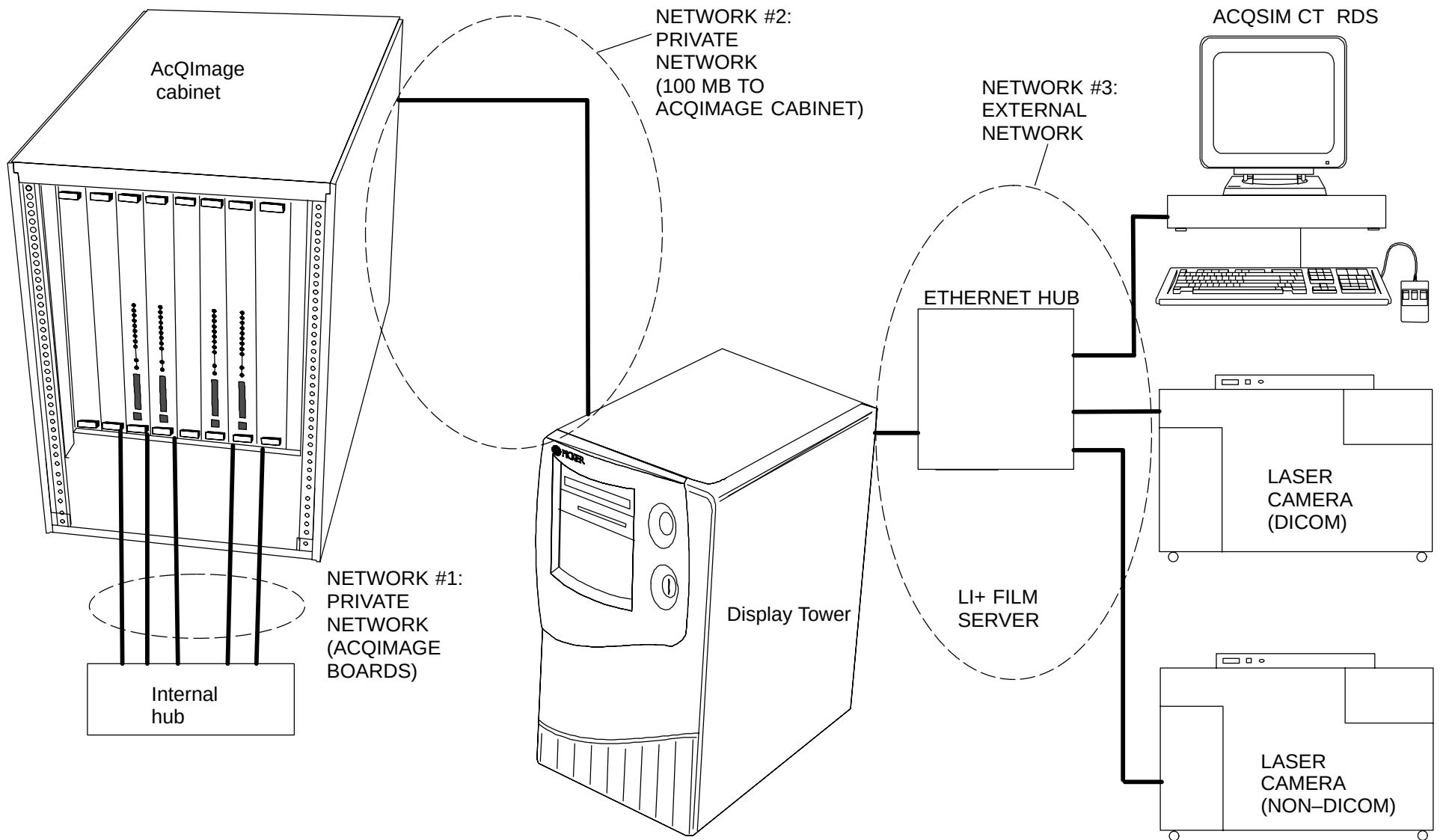


FIGURE 377 EXAMPLE OF PRODUCT'S THREE ETHERNET NETWORKS

NETWORK PORT CONFIGURATION

If there are any Picker option (network) devices or hospital network connections in the installation then configuration of the ethernet NETWORK PORT will be necessary. The NETWORK PORT is not configured as shipped from the factory. Each installation is **unique and must be done on site**. It will be necessary to enter IP address information as well as other parameters for the local DISPLAY CONSOLE and other devices connected to it via the network. This section refers to configuration details for the DISPLAY CONSOLE (and will also work for the UltraZ Diagnostic Workstation) . As a rule all other devices connected on the network will need to be configured locally on their own as well. Refer to individual installation manuals and instructions for these specific devices (e.g. UltraZ Diagnostic Workstation, Voxel-Q, Li +, Dicom Camera, etc.).

Values are entered in the Configure | Network/Camera option of the Service Application.

The Network/Camera Configure Option consists of *Local Parameters* and *Other Devices* tabs:

Network/Camera (Local Parameters): Allows complete local Network configuration:

Network Configuration for this Computer	
Host Name:	
IP Address:	
Aliases:	
Net Mask:	
Default Router:	

These parameters define how other computes in the network communicate with this computer. The name and address fields are the name and address of this computer on the network. To configure names and addresses of other computers, use the 'General Network' tab.

Camera Configuration for this Computer	
Application Entity Title:	
Connection Timer (secs):	
Packet Timer (secs):	

These parameters are used when communicating with cameras. The Application Entity Title is the name of this computer when communicating with DICOM cameras. To configure DICOM or NFP cameras, use the 'Camera Specific' tab.

DICOM Configuration for this Computer	
Application Entity Title:	
PORT:	
Inter PDU Timer (secs.):	
Connection Timer (secs.):	
Idle Timer (secs.):	
Maximum PDU Length:	

These parameters define the defaults used by the DICOM software. The Application Entity Title is the name of this computer when communicating with DICOM cameras. To configure DICOM devices, use the 'DICOM Specific' tab.

FIGURE 378 NETWORK/CAMERA (LOCAL PARAMETERS)

LOCAL PARAMETERS

LOCAL PARAMETERS consists of three subsets, *Network Configuration for this Computer*, *Camera Configuration for this Computer* and *DICOM Configuration for this Computer*. Refer to Figure 378.

Network Configuration for this Computer

Host Name: Used by other devices on the network to communicate with the DISPLAY CONSOLE. Examples of names might be st.mary1, oncology2, picker2077, etc. Names must be limited to 16 characters or less.

- IP Address: Associated with the Host Name. This will be the address through which the DISPLAY CONSOLE will communicate with network devices. If the scanner system and its networked options are contained within a closed system and not connected to any outside hospital network, then you may choose a suitable IP address. You must choose an address that conforms with standard networking conventions (refer to [NETWORK ADDRESSING CONVENTIONS](#) for assistance). If the network is not connected as a closed local network, then you should consult the hospital network administrator or a field service networking specialist for assistance on the assignment of the IP address.
- Aliases: Any alternate names assigned to this host name are entered here. In general no aliases are required.
- Net Mask: A default value of 255.255.255.0 may be entered in case you have a closed network. **You must consult the network administrator or field service networking specialist for information on setting this value if you are connected to a hospital network.
- Default Route: Consult the network administrator or field service networking specialist first for information on setting this value if you are not part of a closed system. For a closed system, the default route should be a valid address for the subnet that is not assigned to any device. A default value equal to the first three fields of the IP address (subnet ID) followed by a 1 in the fourth field would be a good default for a simple closed network. This assumes there is no 1 device on the net. For example, a good default route value for the 192.168.10.xx subnet would be 192.168.10.1 (refer to [NETWORK ADDRESSING CONVENTIONS](#) for assistance).

Camera Configuration for this Computer

This box supplies configuration information for DICOM camera print support supplied by this computer. In the box accept the default timer values. These should not be changed unless instructed to do otherwise by a field specialist. The default title that appears for the Application Entity Title is OK as long as there are no other DICOM devices on the net with the same title. This title represents a logical association with the DICOM process on this computer that supports the camera function. The name chosen for this title is arbitrary in some sense but should be indicative of its functionality for informational purposes. If you rename the title be advised that each Application Entity Title for a device on the net should be set to a unique, 16 character maximum name.

DICOM Configuration for this Computer

This box supplies configuration information for DICOM storage support supplied by this computer. In the box entitled *DICOM Configuration for this Computer*, accept the default timer, port, and PDU Length values. These should not be changed unless instructed to do otherwise by a field specialist. The default title that appears for the Application Entity Title is OK as long as there are no other DICOM devices on the net with the same title. This title represents a logical association with the DICOM process on this computer that supports the storage function. The name chosen for this title is arbitrary in some sense but should be indicative of it's functionality for informational purposes. If you rename the title be advised that each Application Entity Title for a device on the net should be set to a unique, 16 character maximum name.

Network Devices				
Device Name	Type	IP Address	Description	DICOM App. Title
console1	Other	XX.XX.XX.XX		

FIGURE 379 NETWORK/CAMERA (OTHER DEVICES)

OTHER DEVICES

In order that the DISPLAY CONSOLE properly recognize and communicate with other devices on the network it is necessary to provide configuration information on those devices. Selecting the *Other Devices* tab displays the screen summary context seen in Figure 379.

OTHER DEVICES consists of three subsets, *Device Type*, *Network Device Configuration* and either *DICOM Device Configuration*. or *Camera Device Configuration*. This screen (Figure 380) appears when you choose to Add, Modify or Delete a device from the summary context screen (Figure 379).

Device Type

Select the type of device to which you will be making changes (RDS/Scanner, Voxel Q, DICOM Network, DICOM Camera, LI+ Camera or other). The box just below the Network Device Configuration box will change according to the device selected. For example, if you selected RDS, then fill in as follows:

Network Device Configuration

- Device Name: This must match the name given to the RDS in it's own configuration procedure for the device to be recognized.
- IP Address: Must match IP address assigned to the RDS in it's own configuration procedure.
- Device Description: This is the name which you wish the user of the Display Console to see and use when referencing this remote network device. Examples might be rds_plus, rds_office, etc.
- Aliases: Alias names assigned to the RDS. In general, no aliases are required.

If a DICOM device is selected, you will need to supply additional information in the *DICOM Device Configuration* box. If this is a camera or Li+ you will also need to supply additional information.

NOTE

Figure 380 shows the DICOM Device Configuration when RDS/Scanner, Voxel Q or DI-COM Net device types are selected.

Figure 381 shows the Camera Device Configuration when DICOM Cameras or LI+ device types are selected. When "other" is selected, a configuration box does not appear.

DICOM Device Configuration

In this box accept the default values if given. These should not be changed unless you are proficient in setting the values or are instructed to do otherwise by a field specialist. The one exception to this is the *Port* field. It must be configured to match what the Port value that the remote box has configured for itself. It will almost always be Port 104, but it can be changed. Refer to Figure 380.

Also it will be necessary to supply an *Application Entity Title*. This title represents a logical association with the process on that remote box that supports that particular DICOM function. The Application Entity Title needs to match the title that was configured on the remote box always (in this example the remote box being the RDS/Scanner).

When completed, select *Apply Changes* and the entry will be loaded.

Select *Done* and a summary context screen of Network Devices will be displayed again after a valid transaction has been completed. At that point you repeat this process for configuration information on additional devices (Voxel Q, DICOM Camera , etc.) until completed.

DEVICE TYPE					
☐ RDS/Scanner	☐ Voxel Q	☐ DICOM Net	☐ DICOM Camera	☐ LI + Camera	☐ OTHER

Network Device Configuration	
Device Name:	IP Address:
Device Description:	Aliases:

DICOM Device Configuration	
Application Entity Title:	Port:
Max Time Between PDUs:	Connection Wait Time:
Maximum PDU Length:	Idle Wait Time:
☐ Insert Length To End Fields	
Reset To Defaults	

Apply Changes	Cancel Changes	Done
--	---	-------------------------------------

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FIGURE 380 NETWORK/CAMERA, (DICOM DEVICE TYPE)

Camera Device Configuration

Follow a similar approach as the DICOM Device Configuration example to supply the information required. In addition, you must select a camera type by clicking, scrolling and selecting from the list. Refer to Figure 381. For the Li+ camera accept all default information unless you are proficient in setting the values or are instructed to do otherwise by a field specialist. In either case, remember to enter a valid Application Entity Title as previously discussed.

NOTE

Application Entity Title is NOT required for the LI Plus Film Server.

Other

For other devices that do not fall into the previous categories, complete a Network Device Configuration page from the Other device types for each device. Refer to Figure 382.

NOTE

If there are multiple devices of the same type on the network that perform similar functions, then each of those devices must be assigned a unique Application Entity Title. For example, if there were two unique UltraZ Diagnostic Workstation workstations on the network each would need a unique identifier. As an example if the first one had the title DICOM_STORAGE then the second one would need to be alternately named DICOM_STORAGE2 or STORAGE2 or something unique. In this case, you need to add each device separately and complete a separate configuration screen for each device.

DEVICE TYPE	
^ RDS/Scanner	^ Voxel Q
^ DICOM Net	◆ DICOM Camera
^ LI + Camera	^ Other

Network Device Configuration	
Device Name:	IP Address:
Device Description:	Aliases:

Camera Device Configuration	
Application Entity Title:	Port:
Camera Type:	Print Timer Time Out:
Connection Test Level:	Read Timer Time Out:
☐ DICOM Print job SOP Class Supported	Concurrent Connects:
Reset To Defaults	

Apply Changes	Cancel Changes	Done
--	---	-------------------------------------

FIGURE 381 NETWORK/CAMERA, ADD DEVICE (CAMERA DEVICE TYPE)

Device Type	
^ RDS/Scanner	^ Voxel Q
^ DICOM Net	^ DICOM Camera
^ LI + Camera	◆ Other

Network Device Configuration	
Device Name:	IP Address:
Device Description:	Aliases:

Apply Changes

Cancel Changes

Done

FIGURE 382 NETWORK/CAMERA, ADD DEVICE (OTHER DEVICE TYPE)

NETWORK ADDRESSING CONVENTIONS

As shown in Figure 383, the AcQSim CT product requires IP network addresses to be configured for the Display console as well as the networked components of an AcQSim CT system.

In the example shown the display tower needs connected to a RDS, a DICOM laser camera, and a Li Plus server connected to a non DICOM camera.

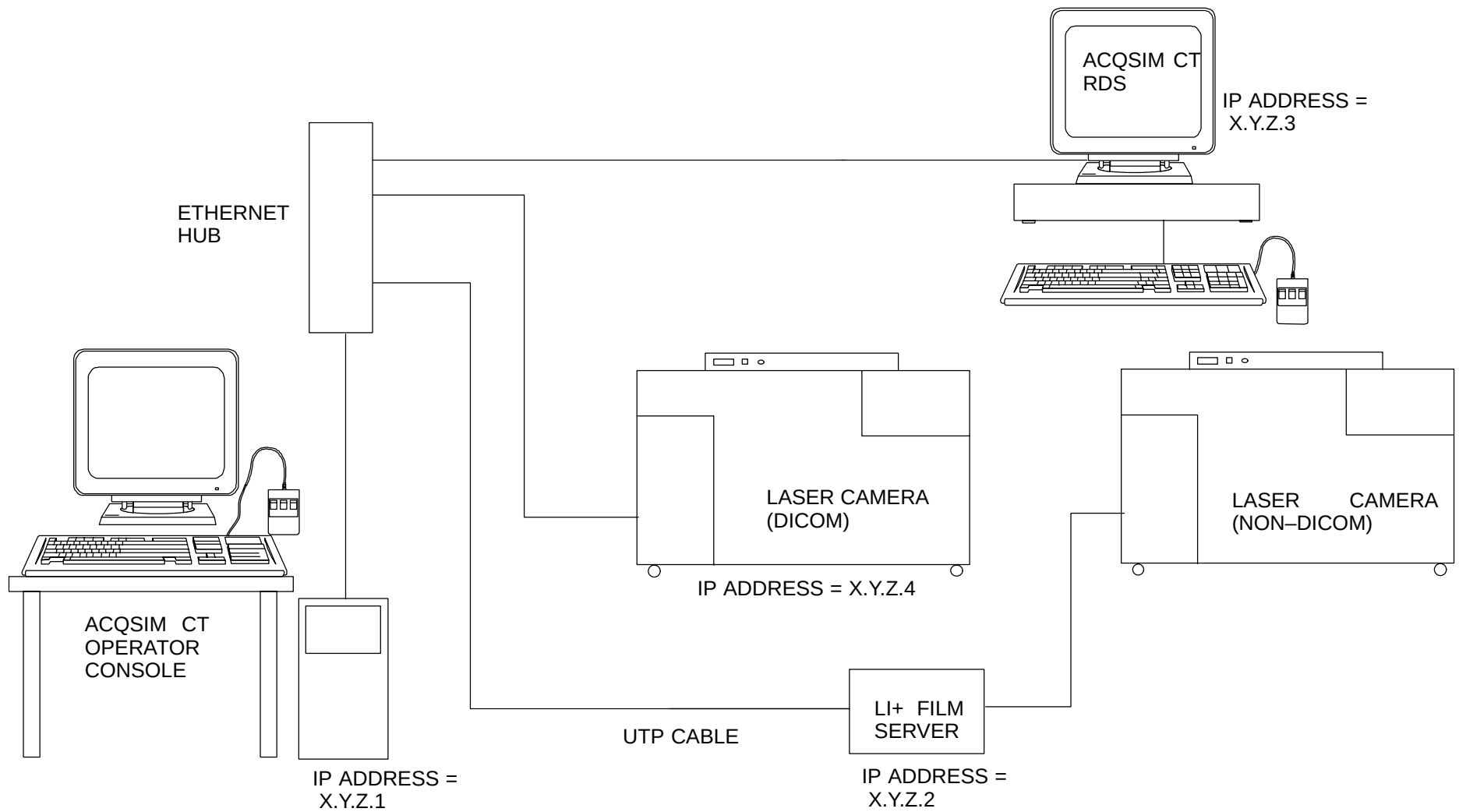


FIGURE 383 AcQSim CT “CLOSED” NETWORK CONFIGURATION

In the example the IP addresses are labeled X.Y.Z.# to indicate that the first three digits of the IP address for each component must be the same when on a closed network. X, Y, and Z can be almost any numbers (See next section).

Figure 383 shows a closed network of only AcQSim CT components and related items. Figure 384 shows a AcQSim CT network connected to another network.

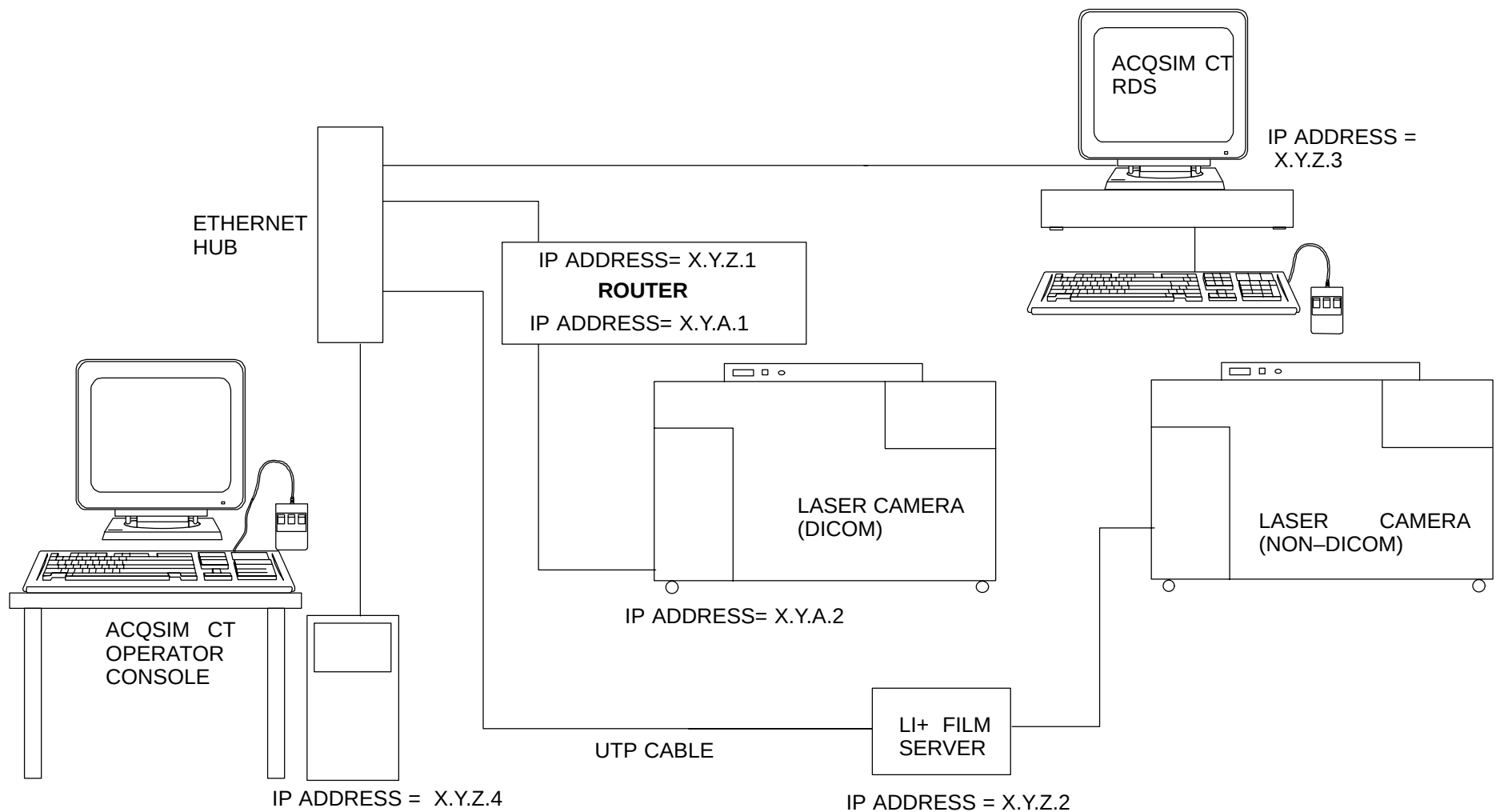


FIGURE 384 AcQSim CT CONNECTED TO ANOTHER NETWORK

In this example (Figure 384) the DICOM laser camera is connected to the X.Y.A network while the AcQSim CT system (and components) is connected to the X.Y.Z network. This example shows the use of a router to connect the two network segments together.

PICKING IP ADDRESSES

If a AcQSim CT system is connected to an external network as shown in the second example, a network administrator or other hospital personnel must provide the IP addresses.

If a AcQSim CT system is connected as closed network, as shown in the first example, and the hospital does not know or care what the IP addresses are, then any IP addresses can be used. The only stipulation is that the address of each node must have the same network ID.

Understanding IP Addresses, Subnetmasks, and Routing

Every node on a TCP/IP network is identified by a unique IP address. This address is used to identify a host on a network; it also specifies routing information on the network. The IP address identifies a computer 32-bit address that is unique across a TCP/IP network. An address is usually represented in dotted decimal notation, which depicts each octet (eight bits, or one byte) of an IP address as its decimal value and separates each octet with a period. For example: 138.238.128.28

NOTE

It is important to note that since an IP address identifies a node on a particular network, each node on the network must be assigned a unique IP address valid for its particular network. Therefore if a machine was assigned an IP address in the 138.238.135 subnet and is subsequently moved into the 138.238.128 subnet, a different IP address must be assigned.

Network ID and Host ID

The IP address contains two pieces of information: the network ID and the host ID for the individual computer.

The network ID identifies a group of computers and other devices that are all located on the same logical network.

The host ID identifies an individual computer within a particular network ID.

IP Address Classes

The Internet community has defined address classes to accommodate networks of varying sizes. Each network class can be discerned from the first octet of its IP address.

The table below shows the relationship between the first octet of a given address and its network ID and host ID fields. It also identifies the total number of network ID's and hosts ID's for each address class in the Internet addressing scheme. In the table a.b.c.d is used to designate the bytes of the IP address.

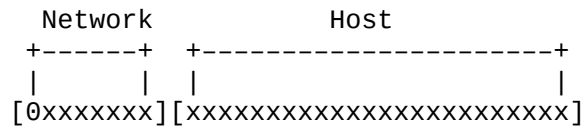
Class	Class Values	Network ID	Host ID	Available Networks	Available Hosts/Nets
A	1–126	a	b.c.d	126	16,777,214
B	128–191	a.b.	c.d	16,384	65,534
C	192–223	a.b.c	d	2,097,151	254

Note that the network addresses 0 and 127 are reserved. Addresses 224 and above are reserved for special protocols and cannot be used as network addresses.

Subnetting

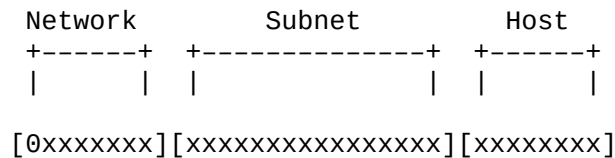
CLASS A SUBNETTING

A class A address could be diagramed as:



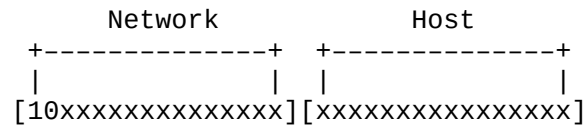
which shows the eight network bits followed by the 24 host bits.

In practice, people don't really attach 16 million hosts to a network so administrators of a Class A site often divide the host address portion into a (sub)network and host portion. (Subnetting is now supported by most operating systems.) Each Class A network number can support up to 65,534 subnets (network numbers 0.0 and 255.255 are reserved) with each having 254 hosts (host numbers 0 and 255 are reserved). This is done by using the 16 high –order bits of the host portion for the subnet number and the lower eight bits for the host as diagramed below.

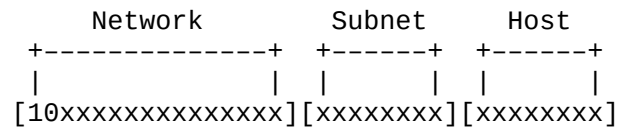


CLASS B SUBNETTING

The first two bits of a Class B address are 1 and 0, the next fourteen bits identify the network and the last sixteen the host, as diagramed:



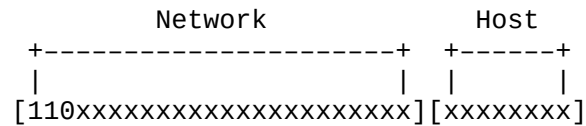
As with the Class A address, we can divide the host portion of a Class B address into subnet and host parts. For instance, let's split our Class B network number on the byte boundary, that is, the eight MSB's of the host portion identifies the subnet and the remaining bits the host, as diagramed:



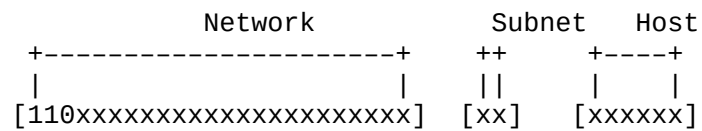
This arrangement allows 254 subnets (0,255 reserved) each with 254 hosts (0,255 reserved).

CLASS C SUBNETTING

The first three bits of a Class C address are 1, 1, and 0, the next 21 bits identify the network and the last eight the host, as diagramed:



Class C networks can be subnetted by using 2 bits of the host address as the subnetwork with 6 bits left for host addresses as diagramed:



This arrangement allows 2 subnets (0,4 reserved) each with 62 hosts (0,63 reserved)

SUBNET MASKS

Subnet masks are 32-bit values that allow the recipient of packets to distinguish the network ID portion of the IP address from the host ID. Like an IP address, the value of a subnet mask is frequently represented in dotted decimal notation.

A subnet mask (or number) is used to determine the number of bits used for the subnet and host portions of the address. The mask is a 32-bit value that uses one-bits for the network and subnet portions and zero-bits for the host portion. For example, below is a Class B address of 191.70.55.130 and apply some different subnet masks. A bitwise logical AND operation is performed between the IP address and the subnet mask as shown:

Here we use a mask that retains the default 16 network and host bits for a Class B address:

191	70	55	130	
1011 1111	1000 0110	0011 0111	1000 0010	IP address
1111 1111	1111 1111	0000 0000	0000 0000	Subnet mask
1011 1111	1000 0110	0000 0000	0000 0000	Result
191	70	0	0	

Employed here is a mask that divides the host portion into a subnet and host that are each eight bits wide:

191	70	55	130	
1011 1111	1000 0110	0011 0111	1000 0010	IP address
1111 1111	1111 1111	1111 1111	0000 0000	Subnet mask
1011 1111	1000 0110	0011 0111	0000 0000	Result
191	70	55	0	

This division allows 254 (256–2 reserved) subnets, each with 254 hosts. This division on a byte boundary makes it easy to determine the subnet and host from the dotted–decimal IP address.

The subnet–host boundary, however, can be at any bit position in the host portion of the IP address. Here, we use a mask that allows more subnets (512–2 reserved), but with the trade–off of fewer hosts (128–2) per subnet:

191	70	55	130	
1011 1111	1000 0110	0011 0111	1000 0010	IP address
1111 1111	1111 1111	1111 1111	1000 0000	Subnet mask
1011 1111	1000 0110	0011 0111	1000 0000	Result
191	70	55	128	

The table below shows default subnet masks for standard IP classes:

Address Class	Default Bits for Subnet Mask	Subnet Mask
Class A	11111111 00000000 00000000 00000000	255.0.0.0
Class B	11111111 11111111 00000000 00000000	255.255.0.0
Class C	11111111 11111111 11111111 00000000	255.255.255.0

NOTE

Not all devices support masking to any bit. Some require that the subnet mask be on byte boundaries only as shown above.

The subnet–host number tradeoff

Here's a table that let's you see at a glance the trade off between the number of subnets and hosts with different subnet masks for both Class B and Class C addresses. We've already subtracted two from the results in the last two columns to take the reserved network and host numbers into account:

Class B Subnetting:

# Mask Bits	Subnet Mask	# Subnets	# Hosts
2	255.255.192.0	2	16382
3	255.255.224.0	6	8190
4	255.255.240.0	14	4094
5	255.255.248.0	30	2046
6	255.255.252.0	62	1022
7	255.255.254.0	126	510
8	255.255.255.0	254	254
9	255.255.255.128	510	126
10	255.255.255.192	1022	62
11	255.255.255.224	2046	30
12	255.255.255.240	4094	14
13	255.255.255.248	8190	6
14	255.255.255.252	16382	2

Class C Subnetting:

# Mask Bits	Subnet Mask	# Subnets	# Hosts
2	255.255.255.192	2	62
3	255.255.255.224	6	30
4	255.255.255.240	14	14
5	255.255.255.248	30	6
6	255.255.255.252	62	2

Routing

The most common implementation of subnetting is placing routers between the subnets. Routers are physical pieces of equipment that connect two or more subnetworks.

A router will have two or more ethernet connections (1 connection per subnetwork) and provides the ability to route network packets to the appropriate subnetwork.

In example 2, the router has connections to the subnetworks X.Y.Z and X.Y.A. The router has an IP address of X.Y.Z.1 for connecting to the X.Y.Z subnetwork as well as an IP address of X.Y.A.1 for connection for the X.Y.A network.

If the AcQSim CT display console sends images to the RDS all of the data is contained in the X.Y.Z network. The router is not used. If the display console sends images to the DICOM camera then the router is used. The router uses the IP address and subnetmask to determine where to forward the network packets.

What's It All Mean?

When installing a AcQSim CT system in a closed network environment and the site does not have a specific IP address required, use a class C address. Do not use .0 or .255 in any of your addressing schemes as they are reserved for special functions. A closed network in this case is one in which the UltraZ and peripherals like camera, UDW, etc are only connected to each other (i.e., no connections exist to the hospital or outside network.)

You should use the following scheme for a closed system:

IP Address	Device
192.168.10.10	Display Console (Scanner)
192.168.10.20	RDS +
192.168.10.30	DICOM Camera
192.168.10.40	Li + Server

and so on

NOTE

The 192.168.XX.XX IP address range is a universally accepted standard for closed networks.

Please note that if the site at a later date wants the equipment connected to a another network, the IP addresses must be changed.

The hospital must provide the IP addresses to use for the AcQSim CT components when connecting the scanner to external networks.



WARNING

Never use the IP addresses 144.54.xx.xx as network addresses as they are reserved for in-house Picker/Marconi networks. Use of these addresses could cause serious network conflicts

CONFIGURE – ACQSIM CT DIAGNOSTIC WORKSTATION

The different CONFIGURE menus for the AcQSim CT Diagnostic Workstation include Site, Network/Camera, Options and Generic configurations. Refer to Figure 385.

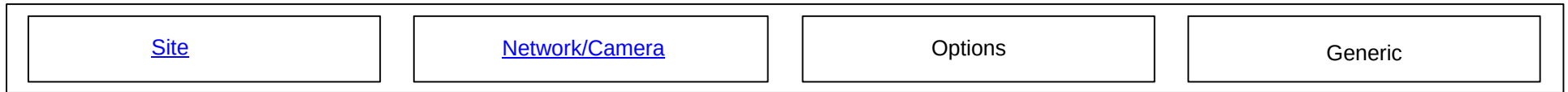


FIGURE 385 CONFIGURATION SELECTIONS

The following procedures detail each individual configuration, based on Figure 385, going from left to right. ** Only **SITE** and **Network/Camera** and **Options** are pertinent for UltraZ Diagnostic Workstation.

You may click on the buttons in Figure 385 to jump to their respective screens.

Site: Allows complete SITE configuration.

Site Configuration

Institution Name

Institution Address

Department Name

Station Name

Equipment Manufacturer

Model Name

Equipment Serial Number

Software Version

Spatial Resolution

Image Compression

Last Calibration Date

Last Calibration Time

FIGURE 386 SITE CONFIGURATION

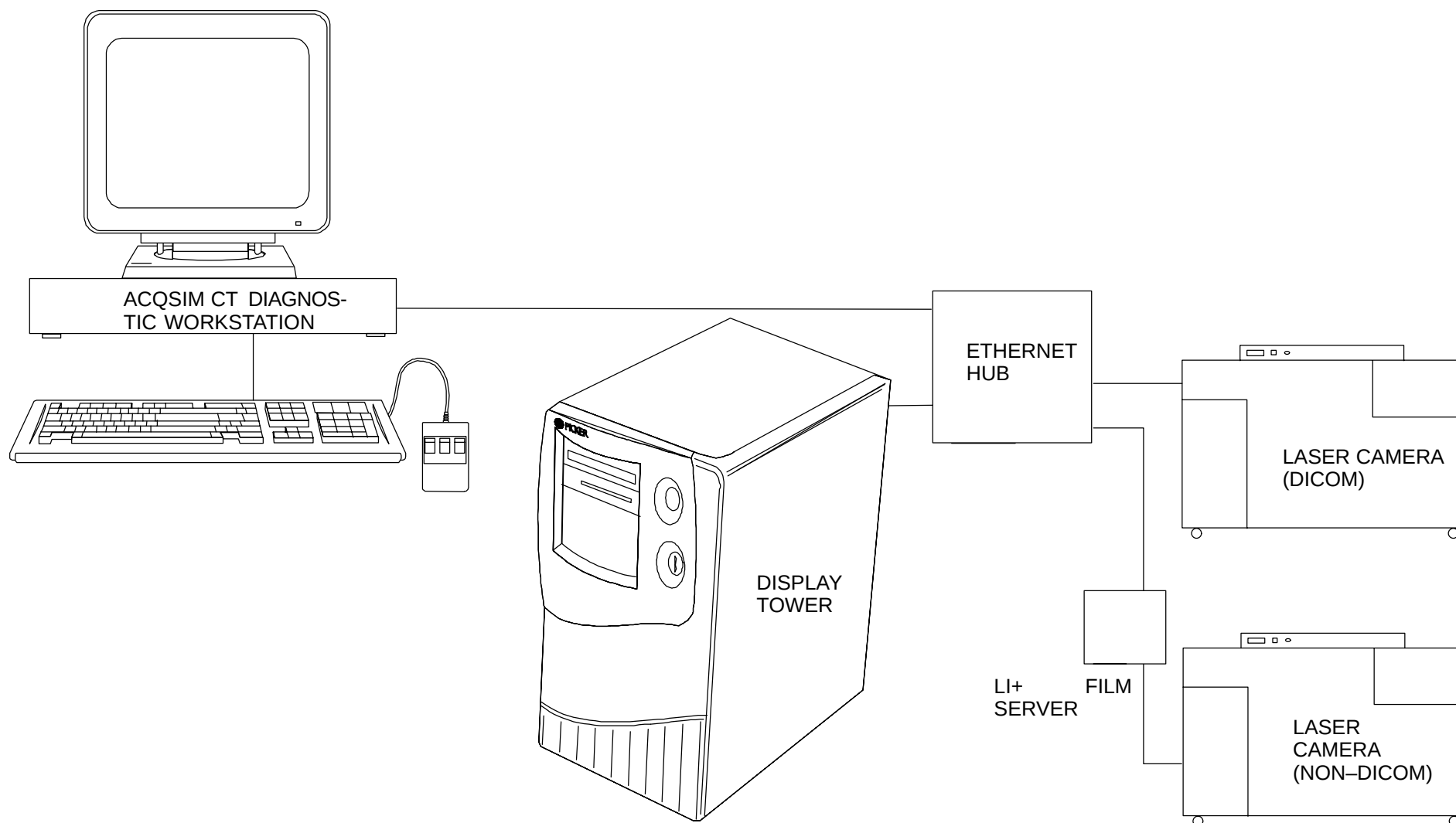


FIGURE 387 EXAMPLE OF ETHERNET NETWORK WITH UltraZ Diagnostic Workstation

NETWORK PORT CONFIGURATION

The UltraZ Diagnostic Workstation is a networked device and configuration must be performed for it to properly communicate with other network devices. Refer to Figure 387 for an example. The NETWORK PORT is not configured as shipped from the factory. Each installation is **unique and must be done on site**. It will be necessary to enter IP address information as well as other parameters for the UltraZ Diagnostic Workstation and other devices connected to it via the network. This section refers to configuration details for the UltraZ Diagnostic Workstation. As a rule, all other devices connected on the network will need to be configured locally on their own as well. Refer to individual installation manuals and instructions for these specific devices (e.g. scanner, other UltraZ Diagnostic Workstation, Voxel-Q, Li +, Dicom Camera, etc.).

Values are entered in the Configure | Network/Camera option of the Service Application.

The Network/Camera Configure Option consists of *Local Parameters* and *Other Devices* tabs:

NOTE

Refer to the Networking Glossary for assistance with definition of terms.

Network/Camera (Local Parameters): Allows complete local Network configuration:

Network Configuration for this Computer

Host Name:

These parameters define how other computes in the network communicate with this computer.

IP Address:

Aliases:

The name and address fields are the name and address of this computer on the network.

Net Mask:

To configure names and addresses of other computers, use the 'General Network' tab.

Default Router:

Camera Configuration for this Computer

Application Entity Title:

These parameters are used when communicating with cameras.

Connection Timer (secs):

The Application Entity Title is the name of this computer when communicating with DICOM cameras.

Packet Timer (secs):

To configure DICOM or NFP cameras, use the 'Camera Specific' tab.

DICOM Configuration for this Computer

Application Entity Title:

These parameters define the defaults used by the DICOM software.

PORT:

Inter PDU Timer secs:

The Application Entity Title is the name of this computer when communicating with DICOM cameras.

Connection Timer secs:

Idle Timer (secs.):

To configure DICOM devices, use the 'DICOM Specific' tab.

Maximum PDU Length:

FIGURE 388 NETWORK/CAMERA (LOCAL PARAMETERS)

LOCAL PARAMETERS

LOCAL PARAMETERS consists of three subsets, *Network Configuration for this Computer*, *Camera Configuration for this Computer* and *DICOM Configuration for this Computer*. Refer to Figure 388.

Network Configuration for this Computer

- Host Name: Used by other devices on the network to communicate with the UltraZ Diagnostic Workstation. Examples of names might be st.mary1, oncology2, picker2077, etc. Names must be limited to 16 characters or less.
- IP Address: Associated with the Host Name. This will be the address through which the UltraZ Diagnostic Workstation will communicate with network devices. If the UltraZ Diagnostic Workstation and networked devices are contained within a closed system and not connected to any outside hospital network, then you may choose a suitable IP address. You must choose an address that conforms with standard networking conventions (refer to NETWORK ADDRESSING CONVENTIONS for assistance). If the network is not connected as a closed local network, then you should consult the hospital network administrator or a field service networking specialist for assistance on the assignment of the IP address.
- Aliases: Any alternate names assigned to this host name are entered here. In general no aliases are required.
- Net Mask: A default value of 255.255.255.0 may be entered in case you have a closed network. **You must consult the network administrator or field service networking specialist for information on setting this value if you are connected to a hospital network.
- Default Route: Consult the network administrator or field service networking specialist first for information on setting this value if you are not part of a closed system. For a closed system, the default route should be a valid address for the subnet that is not assigned to any device. A default value equal to the first three fields of the IP address (subnet ID) followed by a 1 in the fourth field would be a good default for a simple closed network. This assumes there is no 1 device on the net. For example, a good default route value for the 192.168.10.xx subnet would be 192.168.10.1 (refer to NETWORK ADDRESSING CONVENTIONS for assistance).

Camera Configuration for this Computer

This box supplies configuration information for DICOM camera print support supplied by this computer. In the box accept the default timer values. These should not be changed unless instructed to do otherwise by a field specialist. The default title that appears for the Application Entity Title is OK as long as there are no other DICOM devices on the net with the same title. This title represents a logical association with the DICOM process on this computer that supports the camera function. The name chosen for this title is arbitrary in some sense but should be indicative of it's functionality for informational purposes. If you rename the title be advised that each Application Entity Title for a device on the net should be set to a unique, 16 character maximum name.

DICOM Configuration for this Computer

This box supplies configuration information for DICOM storage support supplied by this computer. In the box entitled *DICOM Configuration for this Computer*, accept the default timer, port, and PDU Length values. These should not be changed unless instructed to do otherwise by a field specialist. The default title that appears for the Application Entity Title is OK as long as there are no other DICOM devices on the net with the same title. This title represents a logical association with the DICOM process on this computer that supports the storage function. The name chosen for this title is arbitrary in some sense but should be indicative of it's functionality for informational purposes. If you rename the title be advised that each Application Entity Title for a device on the net should be set to a unique, 16 character maximum name.

Network Devices

Device Name	Type	IP ADDRESS	Description	DICOM App. Title
console1	Other	xx.xx.xx.xx		

Add Device

Modify Selected Device

Delete Selected Device

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FIGURE 389 NETWORK/CAMERA (OTHER DEVICES)

OTHER DEVICES

In order that the UltraZ Diagnostic Workstation properly recognize and communicate with other devices on the network it is necessary to provide configuration information on those devices. Selecting the *Other Devices* tab displays the screen summary context seen in Figure 379.

OTHER DEVICES consists of three subsets, *Device Type*, *Network Device Configuration* and either *DICOM Device Configuration*. or *Camera Device Configuration*. This screen (Figure 390) appears when you choose to Add, Modify or Delete a device from the summary context screen (Figure 389).

Device Type

Select the type on which you will be making changes (RDS/Scanner, Voxel Q, DICOM Network, DICOM Camera, LI+ Camera, other). The box just below the Network Device Configuration box will change according to the device selected, e.g., if you selected RDS/Scanner (for the scanner), then fill in as follows:

Network Device Configuration

- | | |
|---------------------|---|
| Device Name: | This must match the name given to the scanner in it's own configuration procedure for the device to be recognized. |
| IP Address: | Must match IP address assigned to the scanner in it's configuration procedure. |
| Device Description: | This is the name which you wish the user of the UltraZ Diagnostic Workstation to see and use when referencing this remote network device. Examples might be scanner, scanner_oncology, etc. |
| Aliases: | Alias names assigned to the scanner. In general, no aliases are required. |

If a DICOM device is selected, you will need to supply additional information in the *DICOM Device Configuration* box. If this is a camera or Li+ you will also need to supply additional information.

NOTE

Figure 390 shows the DICOM Device Configuration when RDS/Scanner, Voxel Q or DI-COM Net device types are selected.
Figure 391 shows the Camera Device Configuration when DICOM Cameras or LI+ device types are selected. When "other" is selected, a configuration box does not appear.

DICOM Device Configuration

In this box accept the default values if given. These should not be changed unless you are proficient in setting the values or are instructed to do otherwise by a field specialist. The one exception to this is the *Port* field. It must be configured to match what the Port value that the remote box has configured for itself. It will almost always be Port 104, but it can be changed. Refer to Figure 390.

It will be necessary to supply an *Application Entity Title*. This title represents a logical association with the process on that remote box that supports that particular DICOM function. The Application Entity Title needs to match the title that was configured on the remote box (in this example, the remote box being the scanner).

When completed, select *Apply Changes* and the entry will be loaded.

Select *Done* and a summary context screen of Network Devices will be displayed again after a valid transaction has been completed. At that point you repeat this process for configuration information on additional devices (Voxel Q, DICOM Camera , etc.) until completed.

Device Type					
☒ RDS/Scanner	☐ Voxel Q	☐ DICOM Net	☐ DICOM Camera	☐ LI + Camera	☐ Other

Network Device Configuration	
Device Name:	IP Address:
Device Description:	Aliases:

DICOM Device Configuration	
Application Entity Title:	Port:
Max Time Between PDUs:	Connection Wait Time:
Maximum PDU Length:	Idle Wait Time:
☐ Insert Length To End Fields	
Reset To Defaults	

Apply Changes

Cancel Changes

Done

FIGURE 390 NETWORK/CAMERA, ADD DEVICE (DICOM DEVICE TYPE)

Camera Device Configuration

Follow a similar approach as the DICOM Device Configuration example to supply the information required. In addition, you must select a camera type by clicking, scrolling and selecting from the list. Refer to Figure 391. For the Li+ camera accept all default information unless you are proficient in setting the values or are instructed to do otherwise by a field specialist. In either case, remember to enter a valid Application Entity Title as previously discussed.

Other

For other devices that do not fall into the previous categories, complete a Network Device Configuration page from the Other device types for each device. Refer to Figure 392.

NOTE

If there are multiple devices on the network, then each of those devices must be assigned a unique Application Entity Title. For example, if there were two unique UltraZ Diagnostic Workstation workstations on the network each would need a unique identifier. As an example if the first one had the title DICOM_STORAGE then the second one would need to be alternately named DICOM_STORAGE2 or STORAGE2 or something unique. In this case, you need to add each device separately and complete a separate configuration screen for each device.

DEVICE TYPE	
☐ RDS/Scanner	☐ Voxel Q
☐ DICOM Net	☒ DICOM Camera
☐ LI + Camera	☐ Other

Network Device Configuration	
Device Name:	IP Address:
Device Description:	Aliases:

Camera Device Configuration	
Application Entity Title:	PORT:
Camera Type:	Print Timer Time Out:
Connection Test Level:	Read Timer Time Out:
☐ DICOM Print job SOP Class Supported	Concurrent Connects:
Reset To Defaults	

Apply Changes

Cancel Changes

Done

FIGURE 391 NETWORK/CAMERA, ADD DEVICE (CAMERA DEVICE TYPE)

DEVICE TYPE					
☐ RDS/Scanner	☐ Voxel Q	☐ DICOM Net	☐ DICOM Camera	☐ LI + Camera	☒ Other
Network Device Configuration					
Device Name:			IP Address:		
Device Description:			Aliases:		
Apply Changes		Cancel Changes		Done	

FIGURE 392 NETWORK/CAMERA, ADD DEVICE (OTHER DEVICE TYPE)

NETWORKING GLOSSARY

10Base–2	a transmission medium specified by IEEE 802.3 that carries information at rates up to 10Mbps in baseband form using low–cost coaxial cable over distances up to 185 meters (607 ft). Also called thin Ethernet, thinnet, thin coax, or cheapernet.
10Base–5	a transmission medium specified by IEEE 802.3 that carries information at rates up to 10Mbps in baseband form using 50 ohm coaxial cable over distances up to 500 meters (1,640 ft). Also called thick Ethernet or thicknet or thick coax, the cable is commonly referred to as yellow cable. Thick Ethernet cable is typically used as a trunk or backbone path of the network.
10Base–FL	IEEE 802.3 Fiber Optic Ethernet. A fiber optic standard that allows up to 2,000 meters (6,560 ft.) of multimode duplex fiber optic cable in a point–to–point link.
10Base–T	a transmission medium specified by IEEE 802.3 that carries information at rates up to 10Mbps in baseband form using twisted pair conductors. Also called unshielded twisted pair (UTP) wire. Using low cost Level 3 or better UTP wiring, 100 meters (328 ft.) of point–to–point link segments are possible. Uses RJ45 connectors and sometimes 50–pin AMP connectors to a patch panel.
100Base–TX	A particular alternative within the 100Base–TX CSMA/CD proposals before the IEEE 802.3 for a 100 Mb/s Ethernet that specifies two pair of UTP5.
100Base–T	A generic name for 100 Mb/s twisted pair CSMA/CD proposals before the IEEE 802.3. Specific proposals include 100Base–Tx and 00Base–T4.

A

Address	a number uniquely identifying each node in a network.
ARP	(Address Resolution Protocol)  that dynamically maps Internet addresses to physical (hardware) addresses on a LAN.
ATM	(Asynchronous Transfer Mode) a new type of cell switching technology which uses fixed–length packets to transmit data from source to destination. ATM uses fixed–length 53–byte cell–switching to transmit data, voice and video over both LANs and WANs. Also referred to as BISDN and Cell Relay.

AUI (Attachment Unit Interface) the branch cable interface located between a MAU (transceiver) and a DTE (typically a workstation). Includes a 15-pin D-sub connector and sometimes a 15-conductor twisted pair cable. Maximum length is 50 meters (164 ft.).

B

Backbone any network considered to provide interconnection among subnetworks.

Bridge a LAN interconnection device used to link two or more local or remote LANs. Bridges are used extensively in LAN systems to extend their physical dimensions or modify their performance. C

Collision an unwanted condition in which two packets are being transmitted over a medium at the same time, resulting in destruction of the data.

CSMA (Carrier Sense Multiple Access) ⓘ
A station wishing to transmit first senses the medium and transmits only if the medium is idle.

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) – a refinement of CSMA in which a station ceases transmission if it detects a collision.

E

Ethernet the LAN technology that uses CSMA/CD physical access method and 10 Mbps digital transmission. The forerunner of the IEEE 802.3 CSMA/CD standard.

F

FDDI (Fiber Distributed Data Interface) ⓘ
and the topology is a dual-attached, counter-rotating Token Ring. The FDDI protocol has also been adapted to run over traditional copper wires.

FOIRL (Fiber Optic Inter Repeater Link) – A fiber optic signaling method based on the IEEE 802.3 standard governing fiber optics. Allows up to 1,000 meters (3,280 ft.) of multimode duplex fiber optic cable in a point-to-point link.

H

Hub/Repeater the central signal distributor, used in a wiring topology consisting of several point-to-point segments originating from a central point. The term hub is often used interchangeably with the term repeater. Multiport 10BASE-T, 10BASE2, and fiber optic (10BASE-FL, FOIRL) repeaters are considered hubs.

I

IEEE 802.3	a physical layer standard that uses the CSMA/ CD access method on a bus topology LAN.
IEEE 802.5 (Token Ring)	the IEEE committee and its specification that defined a LAN protocol suite. Originated by IBM, now an IEEE standard for a token–passing, ring network that can be configured in a star topology. Token Ring cards are available in 4 Mb/s and 16 Mb/s versions. Subsequent upgrades for fiber are specified in ANSI X3T9.
IP (Internet Protocol)	a connectionless protocol which provides best–effort delivery of datagrams across an internet (the network layer protocol of the TCP/IP protocol suite).
IPX	(Internetwork Packet Exchange) a network layer protocol developed by Novell, Inc. and used in NetWare implementations.

L

	LAN (Local Area Network) – a network system that provides a relatively small area with high–speed data transmission at a low error rate. May include PCs, printers, minicomputers, and mainframes linked by a transmission medium such as a coaxial cable or twisted pair wiring.
Learning bridge	a bridge which automatically learns the topology of the LAN addresses of each node as it receives packets. Requires little or no setup at time of installation.

M

MAU	(medium attachment unit) – a device used to attach a processing node to a network at the physical level. An example is the transceiver used to attach devices to an Ethernet cable.
Medium	a physical conduit for data transmission, e.g., coaxial cable or radio waves.
Multiport repeater	a repeater that collects signals from one transmission channel and, after performing the standard repeater functions, retransmits the signals to more than one new transmission channel.
Multiport transceiver	a transceiver that allows a number of devices to be attached to one LAN transceiver attachment on the backbone network.

N

Node	a point where one or more functional units interconnect transmission lines.
------	---

P

Packet	the basic unit of data transfer in LANs. A chain of one or more buffers that compose a network message.
Packet Internet Groper	(Ping) – A program used to test reachability of destinations by sending them an ICMP echo request and waiting for a reply.
Protocol	the rules or conventions used to govern the exchange of information between networked nodes.
R	
Repeater	a hardware device that regenerates LAN signals to extend the length, topology or interconnectivity of the network, or converts signals between media at the same time.
Rmon	Remote network monitoring probe – a device that was designed to help perform network management on a network segment.
RJ45	this connector is a 10BASE–T standard for connecting UTP cabling. They are inexpensive and easy to install onto UTP cable.
Router	a hardware/software product which receives network layer datagrams and forwards them to their destination based on the network layer address in the packet.
S	
SNMP	(Simple Network Management Protocol) – a set of rules for performing network management functions. Approved for use with TCP/IP in UNIX environments. Created within the Internet community using the RFC process.
Subnetwork	a network that has been connected to a larger and more powerful network system by a bridge or router.
T	
TCP	(Transmission Control Protocol) the transport protocol offering a connection–oriented transport service in the Internet suite of protocols.
TCP/IP	the internetworking protocols developed by the U.S. government's Advanced Research Project Agency (ARPA). Widely adopted and supported by computer and software manufacturers as a standard computer networking protocol.
Thicknet	see 10BASE5

Thinnet – see 10BASE2

Token Ring (IEEE 802.5) the IEEE committee and its specification that defined a LAN protocol suite. Originated by IBM, now an IEEE standard for a token-passing, ring network that can be configured in a star topology. Token Ring cards are available in 4 Mb/s and 16 Mb/s versions. Subsequent upgrades for fiber are specified in ANSI X3T9.

Topology the physical arrangement of devices in a network, regardless of their logical relationships. Types include star, ring, and bus.

Transceiver the attachment hardware connecting the controller interface to the transmission cable in IEEE 802.3 networks.

Twisted pair wiring two insulated wires twisted together and used for transmission (the twisting creates a low level of noise elimination).

U

UTP Unshielded twisted pair – see 10BASE-T

W

WAN (wide area network) – a network that includes nodes distributed over a larger geographic area than can be served by a LAN.

This ends the Network/Camera section of the CONFIGURE option from the main Service Apps menu.

NOTE

The next section displays the AcQImage section of the CONFIGURE option from the main Service Apps menu. The AcQImage selections have defaults set at the factory. The following screens have been provided to show the numerous settings/options.

If you wish to change a setting and are unsure of the terminology, contact Service Engineering.

After selecting the AcQImage button from the CONFIGURE menu. Submenus consist of [Scan](#), General, Recon, Detector, Hardware.

AcQImage (Scan):

Shows the AcQImage Scan configuration. (The opening screen defaults to SCAN:)

Background Flag:	Off ☐	Log Check:	Off ☐
Background Freq.:	BACK6K ☐	Log Check Time:	0
Background Int. Pd:	5243	Rotor Frequency:	0
		Rotor Spin Down Time:	2700
Delta Data Mode:	Off ☐	Rotor Spin Warning:	120
Detector Averaging:	Off ☐		
Digital Check Signal:	Off ☐	Pilot Couch Speed:	100
Check Sig. Frequency:	DIG10K ☐	Pilot Samples:	8
Logger:	Linear ☐	Pilot Column Index Direction:	Source_Fan ☐
Read Time/Count:	Count ☐	Pilot Row Index Direction:	COUCH ☐
Transfer Counter Test:	OFF ☐	Pilot Sampling Direction:	CCW ☐

FIGURE 393 ACQIMAGE CONFIGURATION – SCAN

AcQImage (General): Shows complete AcQImage General configuration:

Termination Token:

☐ FILMNTOFF☐ TAKEBACK
☐ XRAYOFF ☐ FRAMEOFF

XSC Used:

ON

Number Of Overscan Detectors

120

Number of Partial Angle Scan Detectors

0

Number of Detectors Ignored (Partial Angle)

0

Units of Table Position

0.02667

Product Type:

2000+

Hangover Time:

10

Software Release:

PLUS_1.0

Column Index Direction:

X+

Row Index Direction:

Y+

GPU Used:

On

FIGURE 394 ACQIMAGE CONFIGURATION – GENERAL

AcQImage (Recon):

Shows complete AcQImage Recon configuration:

Database Network Host:	localhost	5 Point Conv. Filter Point 1:	0.07291
Database Type Identifier:	1.1.1.1	5 Point Conv. Filter Point 2:	.25
Adaptive Filter Max. Weight:	1	5 Point Conv. Filter Point 3:	0.354167
Adaptive Filter Floor (No Scatter):	20000	9 Point Conv. Filter Point 1:	0.04
Adaptive Filter Floor (Scatter):	20000	9 Point Conv. Filter Point 2:	0.08
Adaptive Filter Low Bound:	27000	9 Point Conv. Filter Point 3:	0.12
Blown Point Threshold:	2000	9 Point Conv. Filter Point 4:	0.16
Isocenter/Detector Dist.:	200	9 Point Conv. Filter Point 5:	0.2
Calcium Correction:	OFF ☐	Moving Average Filter Points:	11
Upper Calmp Value:	38500	Reconstruction Shape:	SQUARE ☐
Order Of Interpolation:	2	Isocenter/Focal Spot Dist.:	640
Output Image Size:	2M	Algorithm Vecotr Source:	RAWDATA ☐
		Projection Comp. Threshold:	2
Raw Data Save Type:	DETECTOR FAN ☐	Reconstruction	
Acquisition Last Step:	NORMAL LAST STEP	Last Step:	Normal Last Step
Acquisition Steps To Skip:	NULL	Recon. Steps to Skip:	NULL
Processing Sequence:	NULL		

FIGURE 395 ACQIMAGE CONFIGURATION – RECON

AcQImage (Detector): Shows complete AcQImage Detector configuration:

Start Of Field Detector:		-1
	Half Field	Full Field
Beam Limiter:	Off ☐	On ☐
Number of Image Generation Detectors	256	512
Start of Gap A:	0	0
Start of Gap B:	0	0
First Detector:	0	0
Detectors in Fan Beam:	0	0
Gap Between Reference/Source Detectors:	1920	0
Number of Reference Detectors	10	20
Number of Scatter Detectors:	18	28
Number of Ripple Threshold:	2048	1024
Maximum Ripple Threshold:	512	256

FIGURE 396 ACQIMAGE CONFIGURATION – DETECTOR

AcQImage (Hardware): Shows complete AcQImage Detector configuration:

AP1 Memory Layout:	0x00003f3f	Disk Type of GAM:	NULL
AP1 603 Memory Amt:	8M	DSP Type for SHUF:	21060
AP1 Cluster Memory Amt:	16M	Ripple Generation DSP Type:	21060
AP1 Cluster 2 memory Amt:	16M	Ripple Correction DSP Type:	21060
AP1 DSP Type:	21060		
		PB1 603 Memory Amt.:	0
CP1 603 Memory Amt:	8M	PB1 Accumulator Memory Size:	4M
		PB1 bitmap for Pipes 32-1:	32
GAM 1 Memory Layout:	0X00000007	PB1 bitmap for Pipes 64-33:	0
GAM 1 603 Memory amt:	8M		
GAM 1 Cluster Memory Amt:	128M		

FIGURE 397 ACQIMAGE CONFIGURATION – **HARDWARE**

Options: Allows complete OPTIONS configuration:

Enabling Key	Option	Expiration
Enabling key strings will be in these locations		
	☐ Picker Service	
	☐ Extra Disks	
	☐ Host ID Change Validation	
	☐ Xray Tube Type: RHINO5_0	
	☐ Auto Send to RDS Support	
	☐ Auto Send to Voxel Q Support	
	☐ MO Drive Attachment	
	☐ Second Tape Drive Attachment	
	☐ Dumb Injectors Support	
	☐ C-Arm Present	
	☐ SVIP Support	
	☐ CTA Support	
	☐ MSSI	
	☐ Pinpoint	
	☐	

FIGURE 398 OPTIONS CONFIGURATION

Generic: Allows _____ configuration:

Generic Configuration File Modification

Load File

Save File

Parameter Selection

Parameter Name	Parameter Value	Comment

Delete Line

Insert Line

Parameter Editing

Parameter Name	Parameter Value	Comment

FIGURE 399 GENERAL CONFIGURATION

Update: After Configuration is complete, click on the “Update Configuration Files” button:

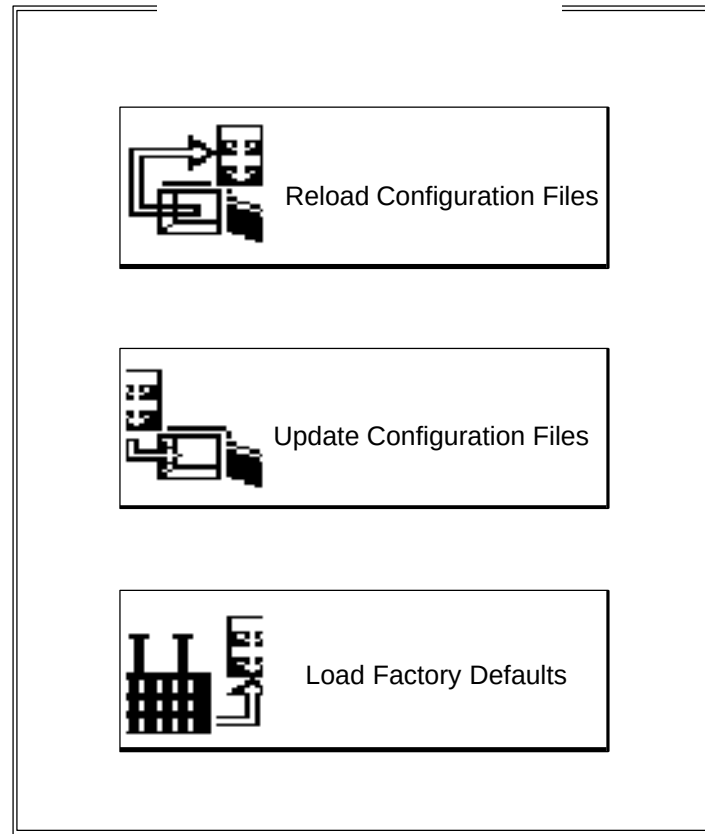


FIGURE 400 UPDATE CONFIGURATION FILES

ERROR LOGGER

Introduction

Errors created during abnormal system operation are recorded in the Error Logger. The contents of the Error Logger can be displayed from the main Service Application pull down menu, called LOG VIEWING.

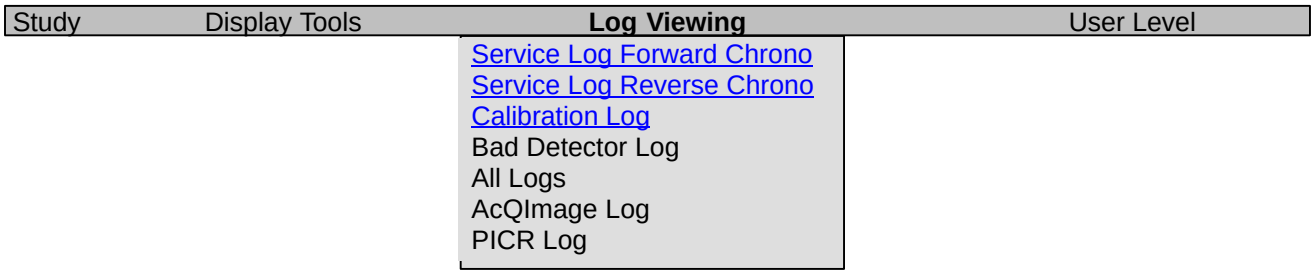


FIGURE 401 LOG VIEWING PULLDOWN MENU

You may select any of the options show in Figure 401. To see samples of the first three error logs, click on the respective heading. Each of the log listing windows allows you to search for specific words or numbers and includes page up, page down and done buttons.

NOTE

The Study and Display Tools pull down menus are covered in the Operator’s manual.


```
-----  
SYSTEM:                                :      0  
FILE:      . ()  
DATE:      /  
INFO:      Scan/Recon failure,Summary error:MODSUMMARY_INTERNAL_ERR,Location  
SYSNAME:
```

```
-----  
SYSTEM:                                :      0  
FILE:      . ()  
DATE:      /  
INFO:      Scan/Recon failure,Summary error:MODSUMMARY_INTERNAL_ERR,Location  
SYSNAME:
```

```
-----  
SYSTEM:                                :      0  
FILE:      . ()  
DATE:      /  
INFO:      Scan/Recon failure,Summary error:MODSUMMARY_INTERNAL_ERR,Location  
SYSNAME:
```

```
-----  
SYSTEM:                                :      0  
FILE:      . ()  
DATE:      /  
INFO:      Scan/Recon failure,Summary error:MODSUMMARY_DEPENDENT_ERR,Locatio  
SYSNAME:
```

```
-----  
SYSTEM:                                :      0  
FILE:      . ()
```

FIGURE 402 SERVICE LOG FORWARD CHRONO – SAMPLE OUTPUT

```

-----
11:12:16  SYSTEM:    SCAN          ERROR:    517   APEID:    0xFFFFFFFF
          FILE:      Errr.cpp (707)
          DATE:      Tue Nov 17, 1998
          INFO:      Bad GSERV state, state is -65536, expected Manual exposure active
          SYSNAME:    bay13-1
          Data Type:  0x00000001

SYSTEM: GSRV      TRACER: ffff8000  PROG:          PC: 0

GSERV STATUS: Bad GSERV state, state is -65536, expected Manual exposure active
GSRV: 8000  SCNS:    0  FUNS:    0  CONT:    1  ROTR:    78  FLMT:    0
XRAY:    0  FLTR:    0  COMP:    0  COLL:   14  LASR:    0  SHUT:    0
SPOT:    1  BLIM:    0   KV:    6   MA:    1  HEAT:    e  FSPD:   190
FPOS: 4afa  CSPD:    0  CHOR: 2040  CVER: 302  TILT:   80  BACK:    0

GPU STATUS: NO ERROR
G4-0:   18  G4-1:   18  G4-2:   80  G4-3:   10  G4-4:  200  G4-5:  400
G4-6:  800  G4-7: ff80  G4-8: 8030  G4-9:    0  G4-A:    0  G4-B: ff00
G4-C:    0  G4-D:    0  G4-E: ff03  G4-F: ff0  G5-0: 1388  G8-0: 5421
G8-1: 997a  G8-2: 8603  G8-4: 2040  G8-5: 302  G8-7: 1080  G8-8: a000
G8-9:    0  G8-A: c7a8  G8-B: 4050  G8-C:    0  G9-0:  ff  MTMP:   19
CTMP:  1d  GPUS:    0  FMOD:    2  FSPD:  190  CMOD:    1  CSPD:    0

SCAN TABLE:
CTYP:          APE: ffffffff          GPUF: ffff  XSCF: ffff  BAC0: ffff
DPPC: ffff  CNTW: ffff  TIMW: ffff  KV: ffff  MA: ffff  FSPD: ffff
XTIM: ffff  HTIM: ffff  RSPD: ffff  SPOT: ffff  FLTR: ffff  COMP: ffff
COLL: ffff  BLIM: ffff  SAMP: ffff  ANGL: ffff  DELY: ffff  CSPD: ffff
ACIN: ffffffff  POIN: ffff  REVB: ffff  INTG: ffff  FDET: ffff
SOF: ffff  EOF: ffff  FAN: ffff  SCAT: ffff  #REV: ffff  GAPA: ffff

```

FIGURE 403 SERVICE LOG REVERSE CHRONO – SAMPLE OUTPUT

```
-----
09:23:12.319  SYSTEM:                               :      1
              FILE:      BadDetBaco.m (1)
              DATE:      Fri Nov 13, 1998
              INFO:      SYSTEM PERFORMANCE: -----
              SYSNAME:   Fri

-----
09:23:12.588  SYSTEM:                               :      1
              FILE:      BadDetBaco.m (1)
              DATE:      Fri Nov 13, 1998
              INFO:      SYSTEM PERFORMANCE: 13-Nov-1998 09:23:12
              SYSNAME:   Fri

-----
09:23:12.600  SYSTEM:                               :      1
              FILE:      BadDetBaco.m (1)
              DATE:      Fri Nov 13, 1998
              INFO:      SYSTEM PERFORMANCE: BACO avg offset: -3.03e-02 v (12111 hz)
              SYSNAME:   Fri

-----
09:23:12.611  SYSTEM:                               :      1
              FILE:      BadDetBaco.m (1)
              DATE:      Fri Nov 13, 1998
              INFO:      SYSTEM PERFORMANCE: BACO max offset: -3.28e-02 v (13120 hz)
              SYSNAME:   Fri

-----
09:23:12.623  SYSTEM:                               :      1
              FILE:      BadDetBaco.m (1)
```

FIGURE 404 CALIBRATION LOG